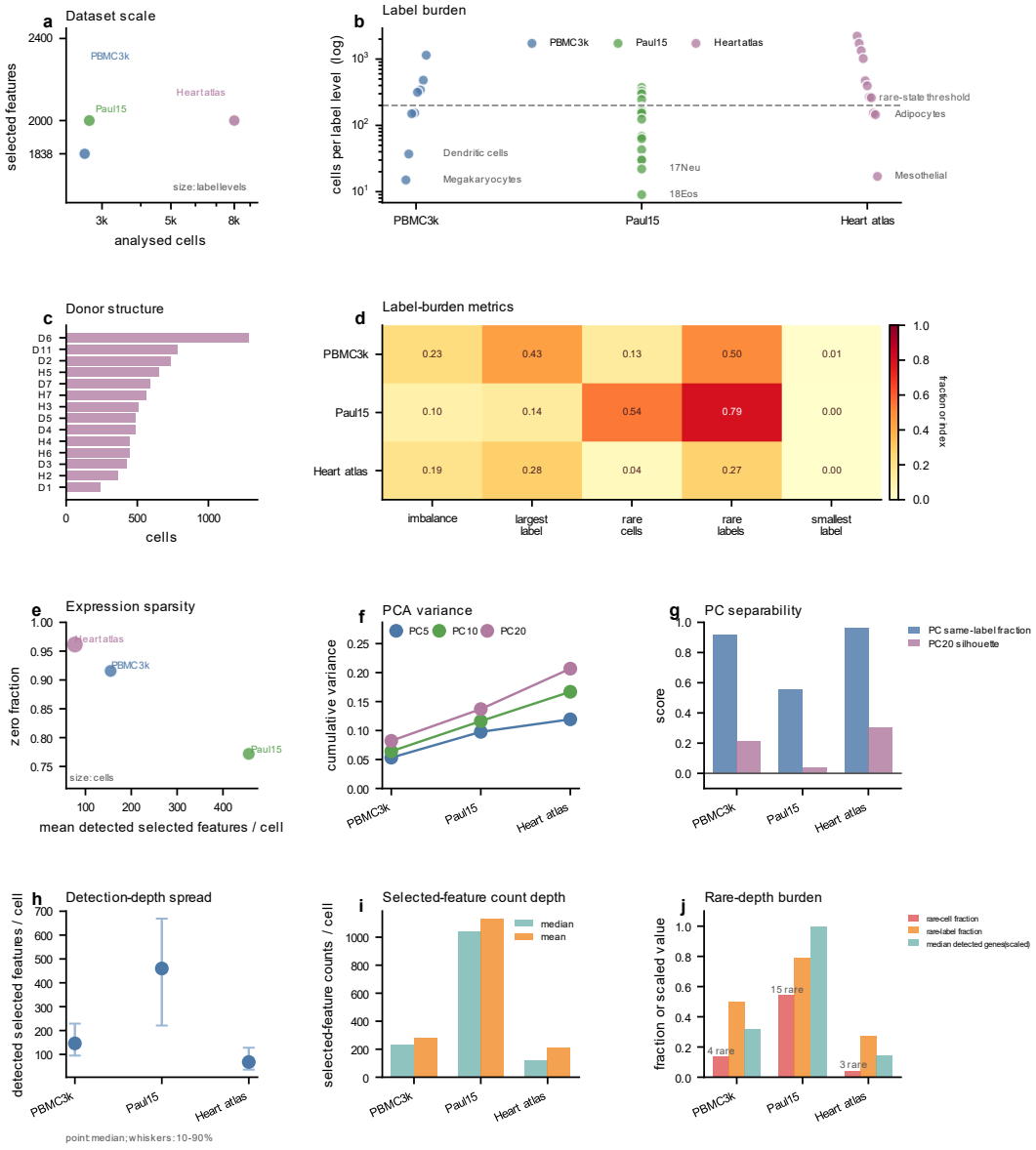
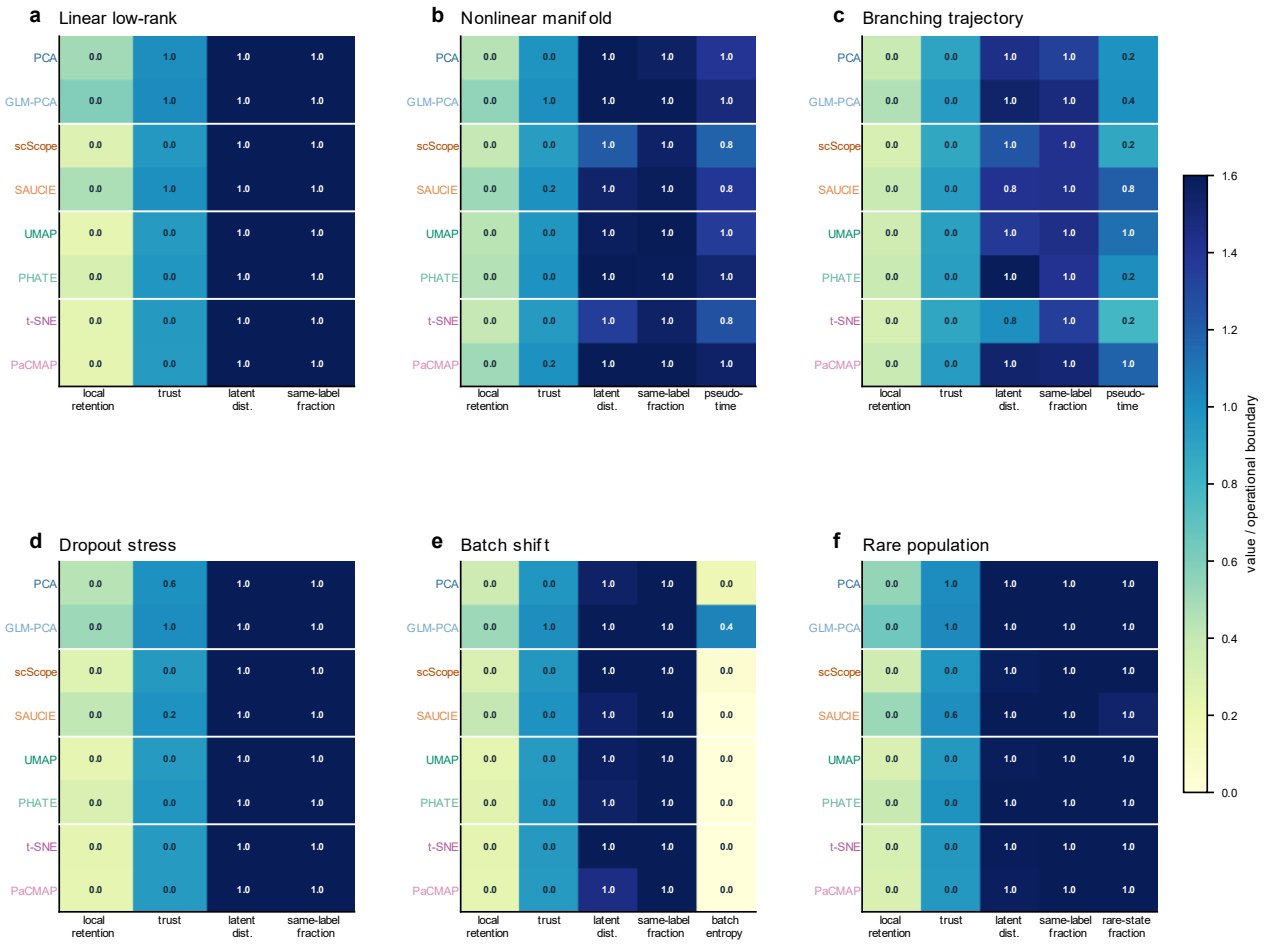
# Supplementary Figures

*Mathematical characterization and empirical evaluation of single-cell transcriptomic dimensionality-reduction methods*

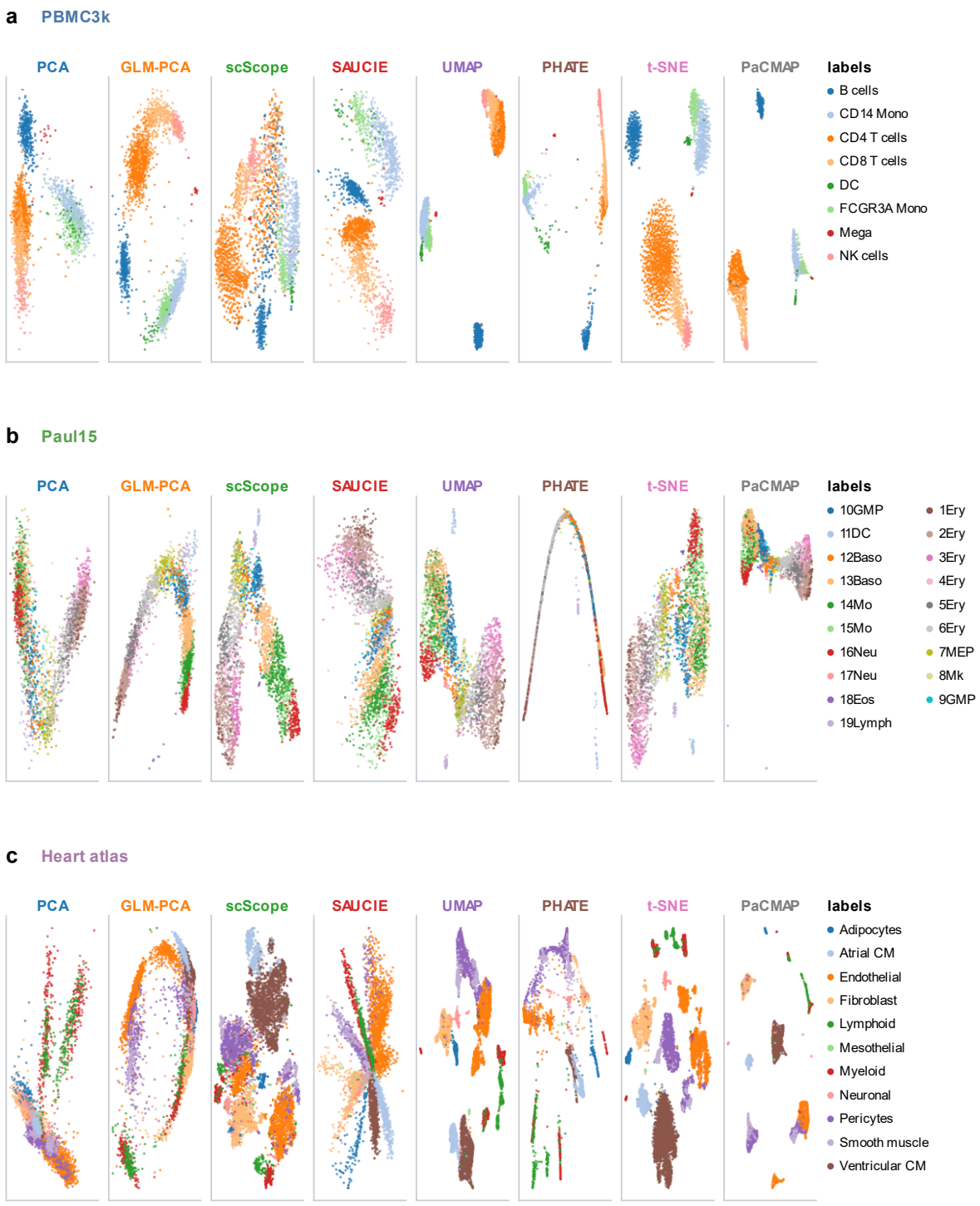
Supplementary Figs. S1-S21



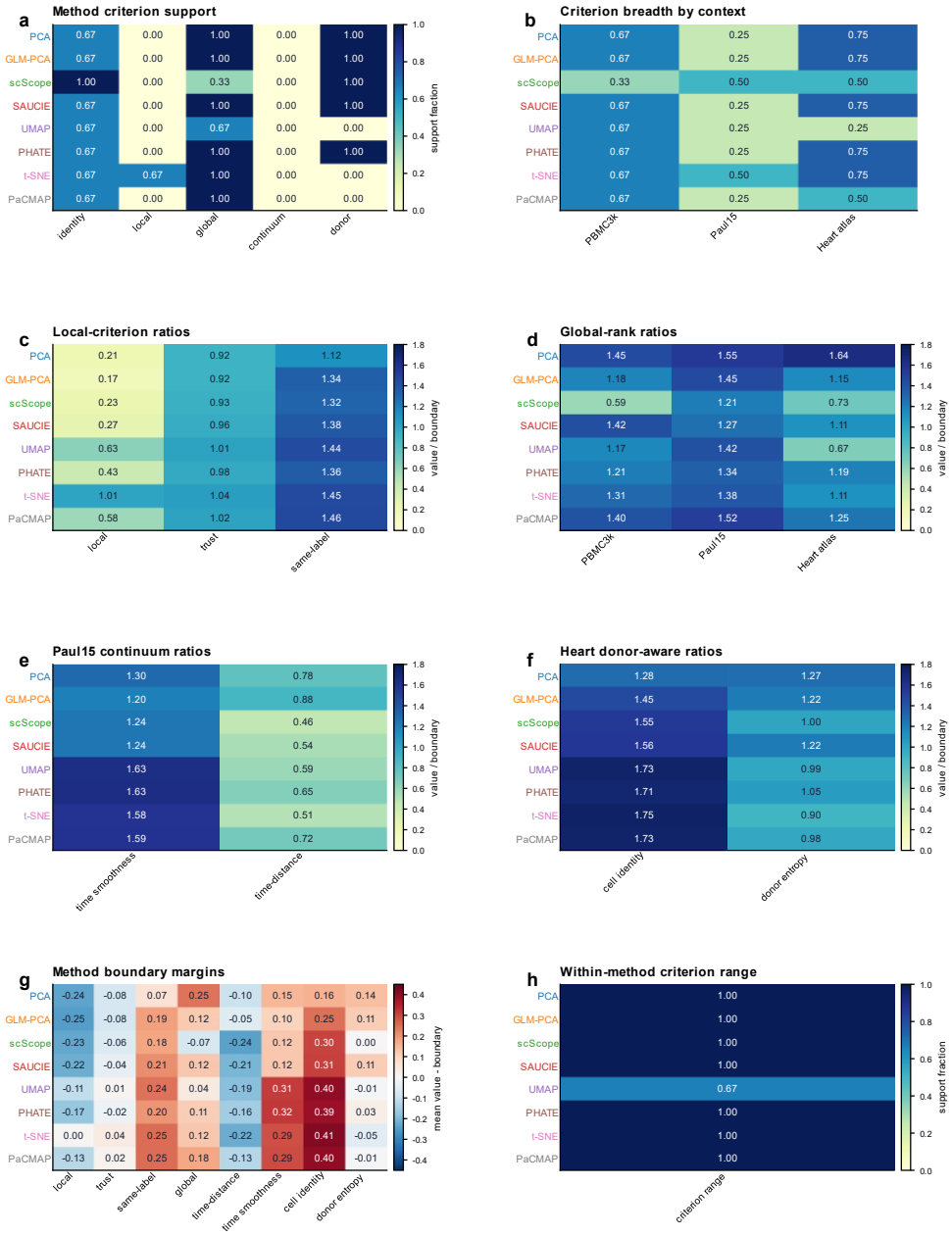

**Supplementary Fig. S1. Dataset structure and analysis context for property-specific embedding interpretation. a,** Dataset scale for PBMC3k, Paul15 and the heart-atlas subset. Points show cells and genes in each processed analysis object; point area denotes the number of annotation levels used in the diagnostic analyses. **b,** Label-size distributions. Each point is one annotation level and the dashed line marks the fixed 200-cell rare-state boundary. **c,** Donor-level cell counts in the heart-atlas subset. **d,** Label-burden metrics, comprising class imbalance, largest-label fraction, rare-cell fraction, rare-label fraction and smallest-label fraction. **e,** Expression sparsity, shown as zero fraction against mean detected selected features per cell; point area denotes cell count. **f,** Cumulative variance explained at PC5, PC10 and PC20. **g,** PCA-reference separability, measured by PC-space same-label neighbour fraction and the PC20 silhouette coefficient. **h,** Detection-depth spread; points are medians and whiskers span the 10th to 90th percentiles of detected selected features per cell. **i,** Median and mean counts per cell within the selected features, not full-library totals. **j,** Rare-state depth burden, combining rare-cell fraction, rare-label fraction and scaled median detected genes; text gives the number of rare annotation levels. These are empirical dataset summaries rather than method-performance results. Here, n denotes datasets, annotation levels, donors or cells as specified by the panel. Public dataset sources, analysed objects, diagnostic criteria and biological definitions are reported in Supplementary Tables S1, S2, S4 and S5; figure-level numerical records and plotting code are available in the public repository.



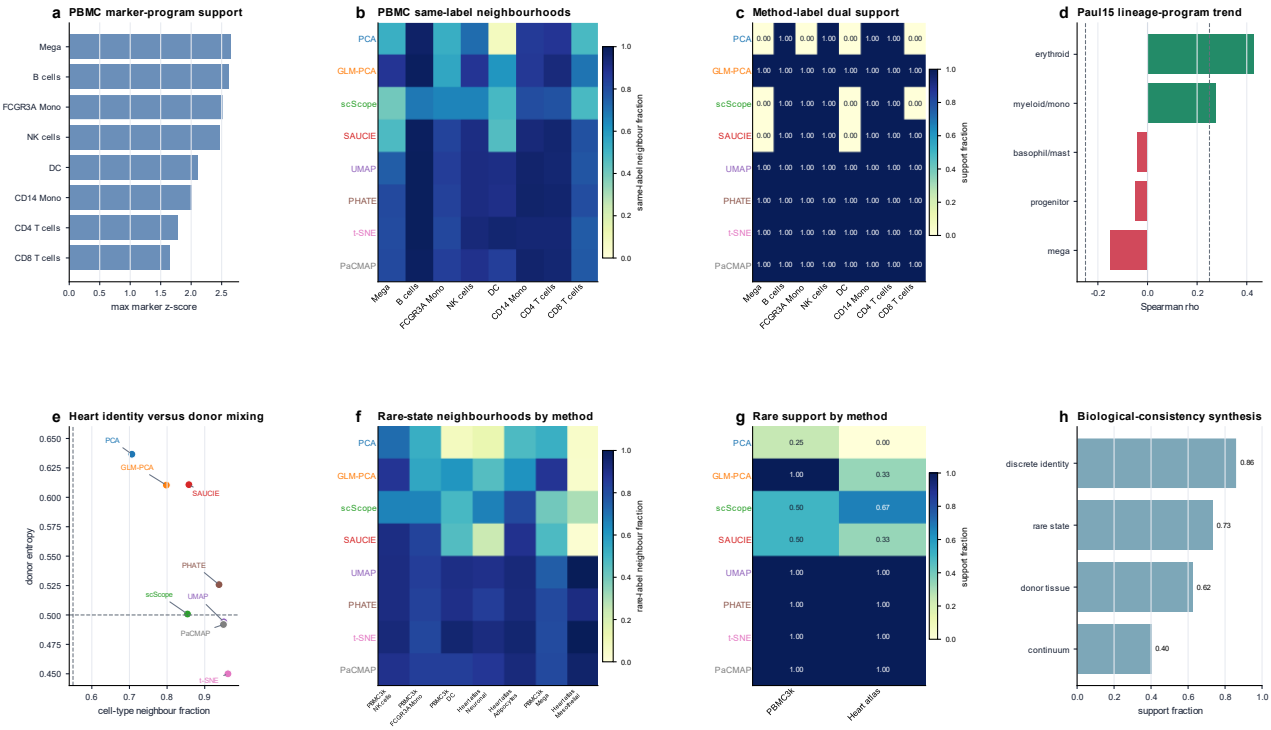

**Supplementary Fig. S2. Metric-level summary of the eight-method controlled stress-test suite. a-f,** Scenarios with known generating structure designed to examine property-specific vulnerabilities. Panels show linear low-rank structure, nonlinear manifold structure, branching trajectory, dropout stress, batch shift and rare-population scenarios. Rows are PCA, GLM-PCA, scScope, SAUCIE, UMAP, PHATE, t-SNE and PaCMAP. Columns show scenario-relevant diagnostics, including local retention against the generating structure, trustworthiness, latent-distance correlation, same-label neighbour fraction, pseudotime-distance correlation, batch entropy and rare-label neighbour fraction. Only metrics defined and calculated for a scenario are displayed. Colour encodes the observed value divided by its fixed operational boundary, capped at 1.6 for display; in-cell numbers give the fraction of five independently generated datasets meeting the boundary. These panels are controlled computational results and do not represent new empirical biology or a comprehensive method benchmark. Operational criteria are reported in Supplementary Table S4, and complete method-scenario-generated-dataset results are reported in Supplementary Table S13.



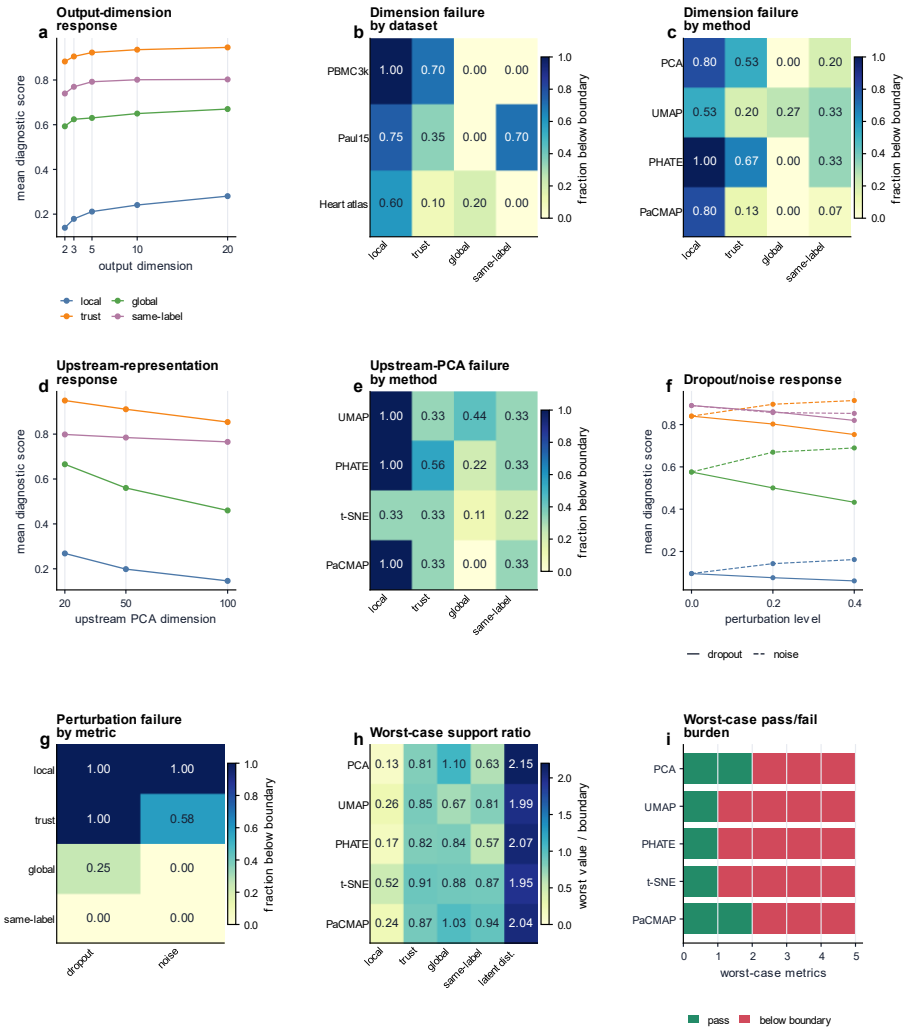

**Supplementary Fig. S3. Full embedding atlas across empirical interpretation contexts. a-c,** Low-dimensional coordinates for the eight-anchor empirical panel in PBMC3k, Paul15 and the heart-atlas subset. Columns show PCA, GLM-PCA, scScope, SAUCIE, UMAP, PHATE, t-SNE and PaCMAP in the same fixed order used throughout the study. Points are cells and colours denote the dataset-specific annotation labels shown at the right of each row. Method-name colours are navigational identifiers only and do not encode a method class or performance level. These maps document the fitted coordinates underlying the matched comparisons; visual separation is not used as a quantitative result. Diagnostics and biological-consistency checks are reported in Figs 3-5. Here, n is the number of cells in the processed analysis object for each dataset-method map; fitted coordinates and plotting code are available in the public repository.



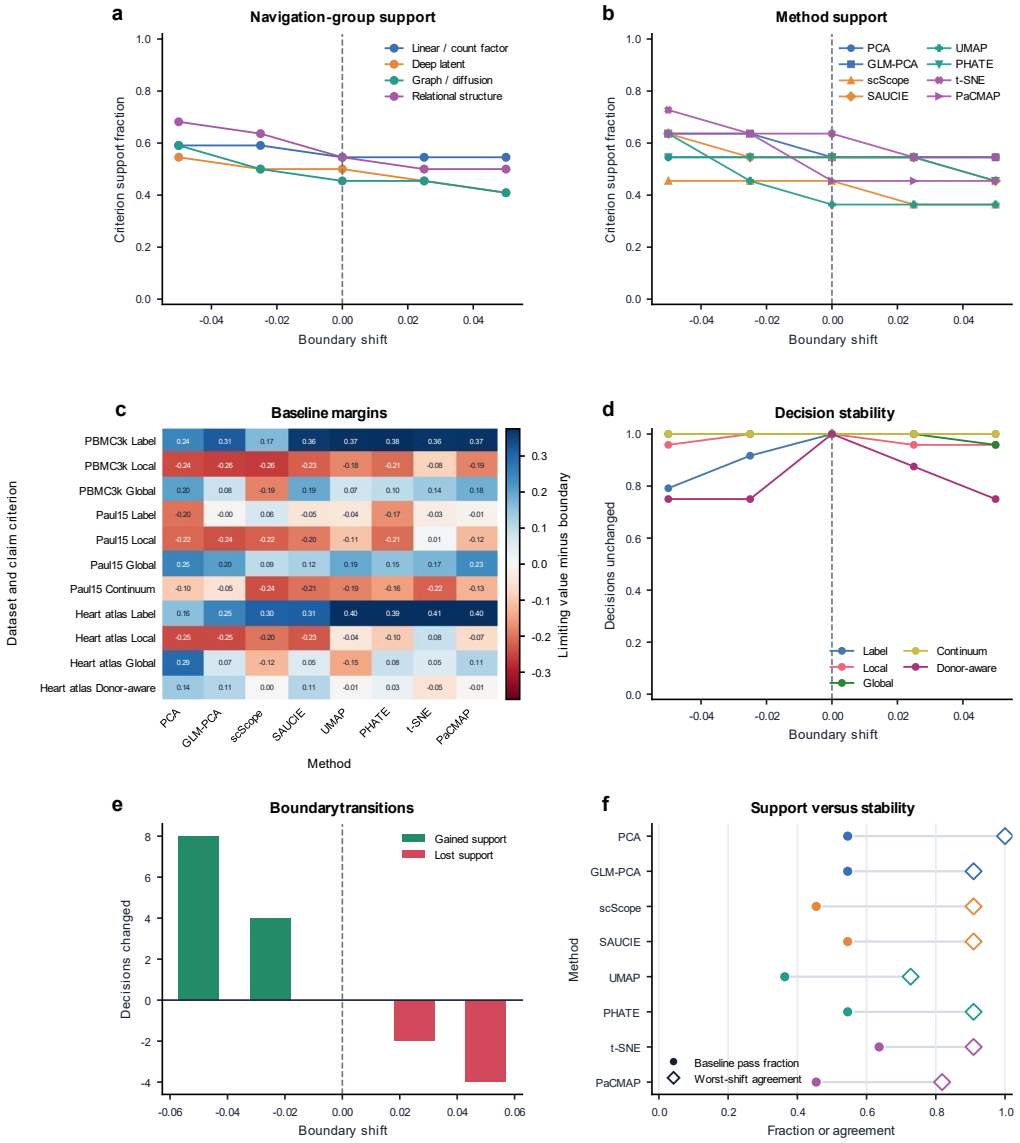

**Supplementary Fig. S4. Diagnostic support, component ratios and boundary margins. a,** Method-level support fractions for label, local-neighbourhood, global-geometry, continuum and donor-aware criteria. **b,** Fraction of applicable criteria supported by each method in PBMC3k, Paul15 and the heart-atlas subset. **c,** Local-criterion component ratios, calculated as the observed value divided by its fixed operational boundary for local retention, trustworthiness and same-label neighbour fraction. **d,** Global-rank ratios across the three empirical contexts. **e,** Paul15 continuum ratios for local pseudotime retention and pseudotime-distance agreement. **f,** Heart-atlas donor-aware ratios for cell-type neighbour fraction and normalised donor entropy. **g,** Mean signed boundary margin for each method and diagnostic component, calculated as observed value minus its boundary. **h,** Within-method criterion range, calculated as the maximum minus minimum valid interpretation-specific support fraction. Heatmap colour denotes support fraction, value-to-boundary ratio or signed margin as specified by each colour bar. Panels c-f share the ratio range 0-1.8; a ratio of one marks the operational boundary. Method-name colours identify methods only. Fractions use valid method-context or method-interpretation rows as denominators; unavailable criteria are excluded rather than counted as failures. Diagnostic definitions and boundaries are reported in Supplementary Table S4; panel-level data and plotting code are available in the public repository.



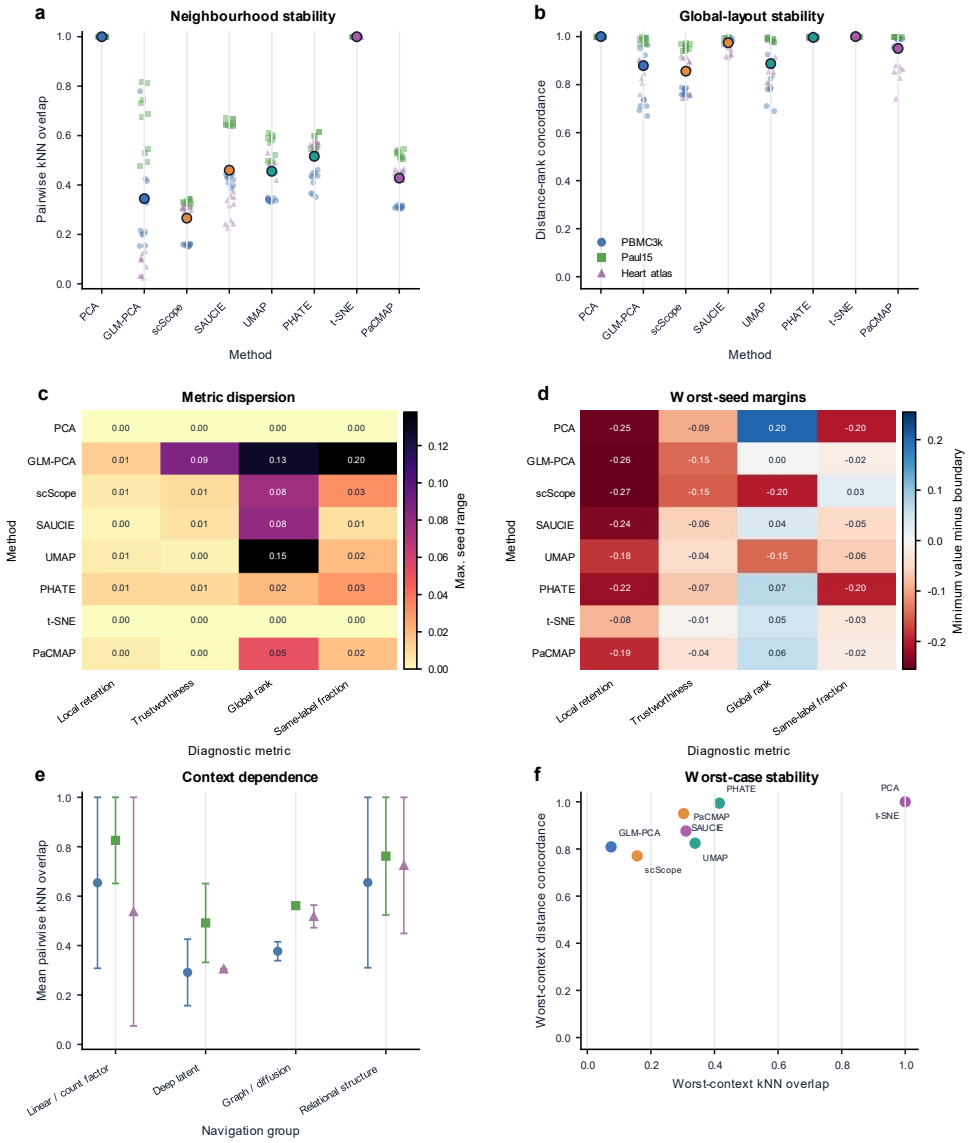

**Supplementary Fig. S5. Expression- and metadata-based consistency checks for embedding interpretation.** **a,** Maximum marker-program z score for each PBMC3k annotation. **b,** PBMC3k same-label neighbour fraction by method and label. **c,** Binary method-label support requiring both the marker-program and neighbour criteria. **d,** Spearman correlation between Paul15 lineage-program score and reference pseudotime; dashed lines mark the fixed absolute-correlation boundary of 0.25. **e,** Heart-atlas cell-type neighbour fraction plotted against normalised donor entropy; dashed lines mark the corresponding diagnostic boundaries. **f,** Rare-label neighbour fraction by method for four PBMC3k and three heart-atlas low-frequency labels. **g,** Fraction of rare labels meeting the neighbour-fraction boundary for each method and dataset. **h,** Descriptive support fractions across discrete identity, rare-state, donor-variable tissue and continuum checks. Method-name colours are identifiers only. Marker programs, pseudotime-associated expression trends and donor metadata were defined without using the displayed embedding coordinates, but they are consistency checks rather than independent experimental validation. Here, n denotes method-label, method-rare-label, lineage-program or method rows. Diagnostic and biological definitions are reported in Supplementary Tables S4 and S5; panel-level data and plotting code are available in the public repository.



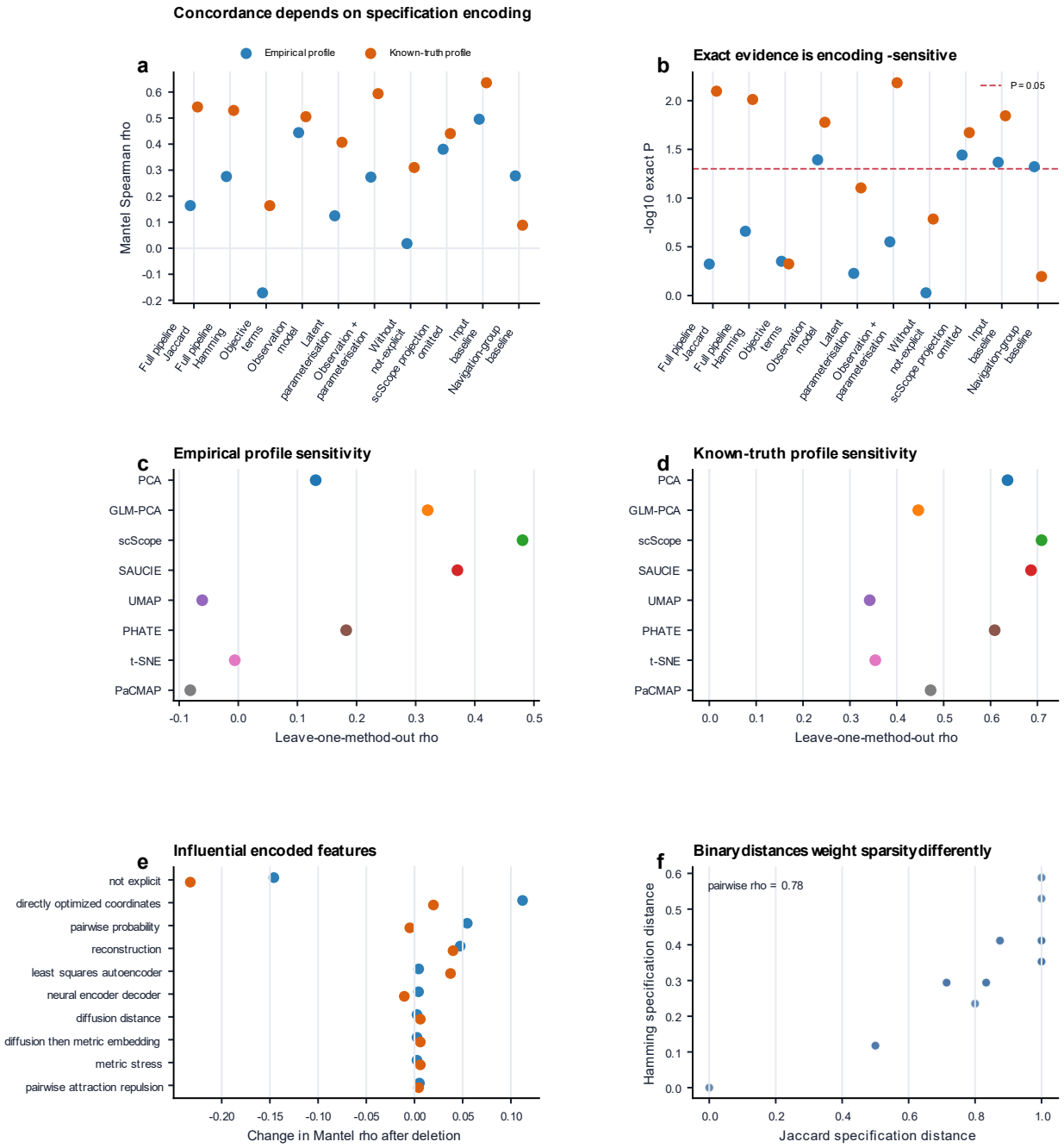

**Supplementary Fig. S6. Robustness responses across dimensionality, upstream representation and perturbation. a,** Mean diagnostic response across output dimensions for local retention, trustworthiness, global-rank correlation and same-label neighbour fraction. **b,** Failure fractions under output-dimension variation by dataset and endpoint. Output dimensions 2, 3, 5, 10 and 20 contribute 20 rows per dataset-diagnostic and 15 per method-diagnostic. **c,** Failure fractions under output-dimension variation by recomputed method and endpoint. **d,** Mean diagnostic response across upstream PCA dimensions for UMAP, PHATE, t-SNE and PaCMAP. **e,** Failure fractions under upstream-PCA variation by method and endpoint. **f,** Mean diagnostic response under dropout and noise perturbations; solid lines denote dropout and dashed lines denote noise. Panels f and g use PBMC3k only, with 2,638 cells, four methods and two computational repeats. Dropout is a non-zero-count masking probability; noise is a multiplier of each gene's baseline log-count standard deviation. **g,** Perturbation failure fraction by endpoint and perturbation type. **h,** Worst-case support ratio for the recomputable subset, calculated as the minimum observed value divided by its fixed operational boundary across the tested conditions. The four non-latent diagnostics pool 28 conditions for PCA, 37 each for UMAP, PHATE and PaCMAP, and 10 for t-SNE. The pools include one auxiliary 1,600-cell simulation; latent-distance correlation uses only this single generated dataset, not the six-scenario replicated suite. The colour scale spans 0-2.2. **i,** Worst-case pass and below-boundary burden by method for the same subset. Curve colours identify diagnostics; heatmap colours encode failure fractions or worst-case ratios, and bar colours distinguish pass from below-boundary outcomes. Mean-response curves do not show uncertainty intervals. Panels summarise computational sensitivity analyses, not independent biological replicates. Diagnostic criteria and complete matched robustness records are reported in Supplementary Tables S4 and S12; panel-level data and plotting code are available in the public repository.



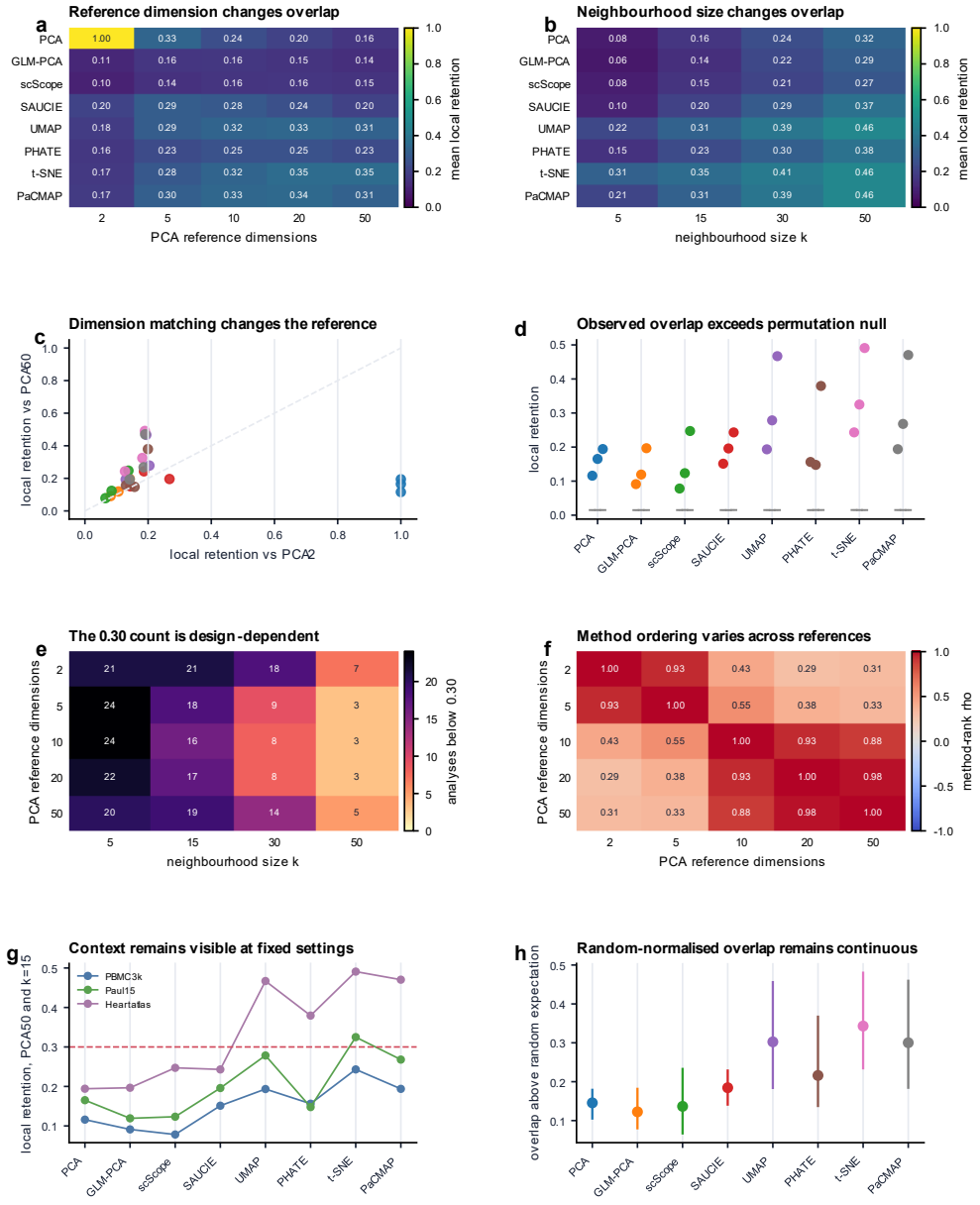

**Supplementary Fig. S7. Interpretation criteria are tested across a defined range of operational boundaries.** **a,** Fraction of applicable decisions meeting the shifted boundary within each method group. Each point summarises 22 method-context decisions from the two empirical-panel methods in that group. **b,** Criterion pass fraction for each method across its 11 applicable dataset-interpretation combinations. Colours identify method groups and symbols identify methods. **c,** Baseline limiting margin for all 88 method-context decisions. Rows are the 11 applicable dataset-interpretation combinations and columns are methods. Each cell is the minimum observed-minus-boundary value among the required components; red denotes a limiting component below its boundary and blue denotes a value above it. **d,** Fraction of decisions unchanged from the unshifted result for label support, local neighbourhoods, global geometry, continuum and donor-aware interpretation. Denominators are 24, 24, 24, 8 and 8 decisions, respectively. **e,** Number of decisions that gained or lost support relative to baseline among all 88 decisions. **f,** Method-level baseline pass fraction and worst-shift decision agreement. Filled circles show the baseline fraction over 11 applicable decisions, open diamonds show the minimum unchanged fraction across the four non-zero shifts and connecting lines aid within-method comparison. Every continuous component boundary was shifted jointly by -0.05, -0.025, 0, 0.025 or 0.05; the vertical dashed line denotes the baseline. These are deterministic sensitivity calculations, not independent replicates, and no confidence intervals or inferential P values are assigned. Component values, boundaries, decisions and denominators are reported in Supplementary Tables S4 and S7.



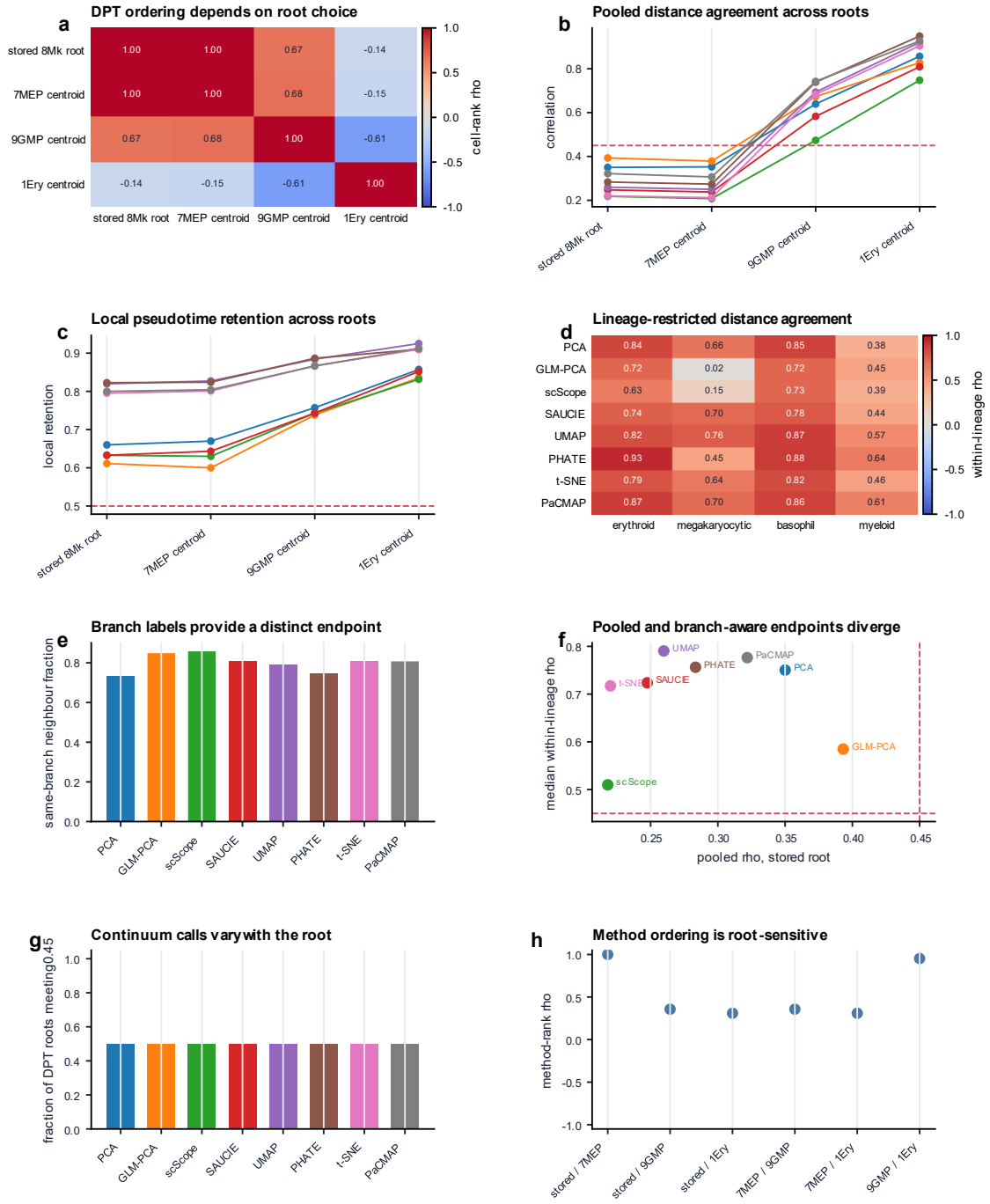

**Supplementary Fig. S8. Local and global embedding structures show distinct random-seed stability.** All eight methods were fitted and scored on the complete PBMC3k, Paul15 and heart-atlas analysis objects, containing 2,638, 2,730 and 8,000 cells, with seeds 0-4. **a,** Pairwise k-nearest-neighbour overlap at k=15. Small points are the ten seed-pair comparisons for each dataset and method; circles, squares and triangles denote PBMC3k, Paul15 and heart atlas. Large outlined circles show the mean across 30 seed-pair comparisons per method, with colour identifying its navigation group. **b,** Pairwise Spearman concordance of embedding-distance ranks over 5,000 fixed cell pairs, using the same point and colour conventions. **c,** Maximum diagnostic range across five seeds, taking the largest range among the three datasets for each method. **d,** Worst seed and context boundary margin, calculated as the minimum observed-minus-boundary value over 15 method-dataset-seed runs. **e,** Context-specific mean pairwise neighbourhood overlap by navigation group. Points average the two methods in each group and whiskers span their observed minimum and maximum; whiskers are descriptive ranges, not confidence intervals. **f,** Worst-context stability for each method. Coordinates are the minima of the three context-specific means for neighbourhood overlap and distance-rank concordance. PCA and the evaluated Scanpy t-SNE configuration produced identical coordinates across five seeds in the recorded software stack; this execution-specific result does not imply general determinism. Seed pairs are computational repeats on the same cells and are not biological replicates. Run-level metrics, comparisons, decisions and software settings are reported in Supplementary Tables S6 and S8.



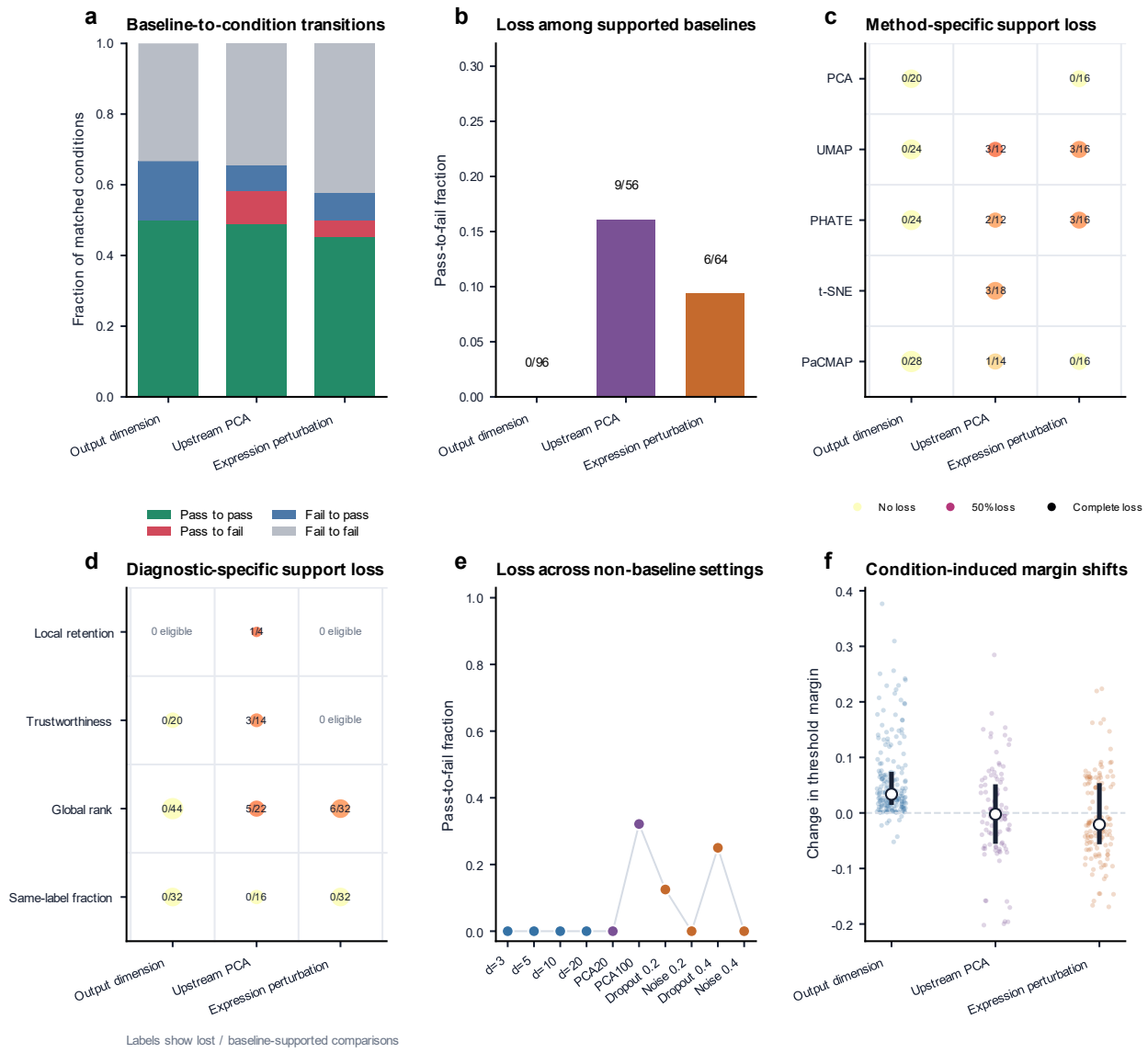

**Supplementary Fig. S9. Specification-to-profile concordance depends on encoding and method composition.** **a,** Exact Mantel Spearman correlation between ten alternative specification encodings and the empirical or controlled-scenario profile. Each point uses eight methods and 28 unique method pairs. **b,** Negative base-10 logarithm of the corresponding exact two-sided P value; the dashed line marks unadjusted P = 0.05. Alternative encodings are exploratory and were not selected by these P values. **c, d,** Leave-one-method-out Mantel correlations for empirical and controlled-scenario profiles. Each point uses seven methods, 21 pairs and all 5,040 method-label permutations. **e,** Change in Mantel correlation after deleting one encoded feature from the full specification; the ten features with the largest absolute change across the two profiles are shown. **f,** Jaccard against Hamming distance for the 28 method pairs under the same 17-feature encoding. The displayed Spearman correlation describes the pairwise relation and is not an independent-pair significance test. Blue and orange identify empirical and controlled-scenario profiles. Complete encodings and exact permutation results are reported in Supplementary Table S9.



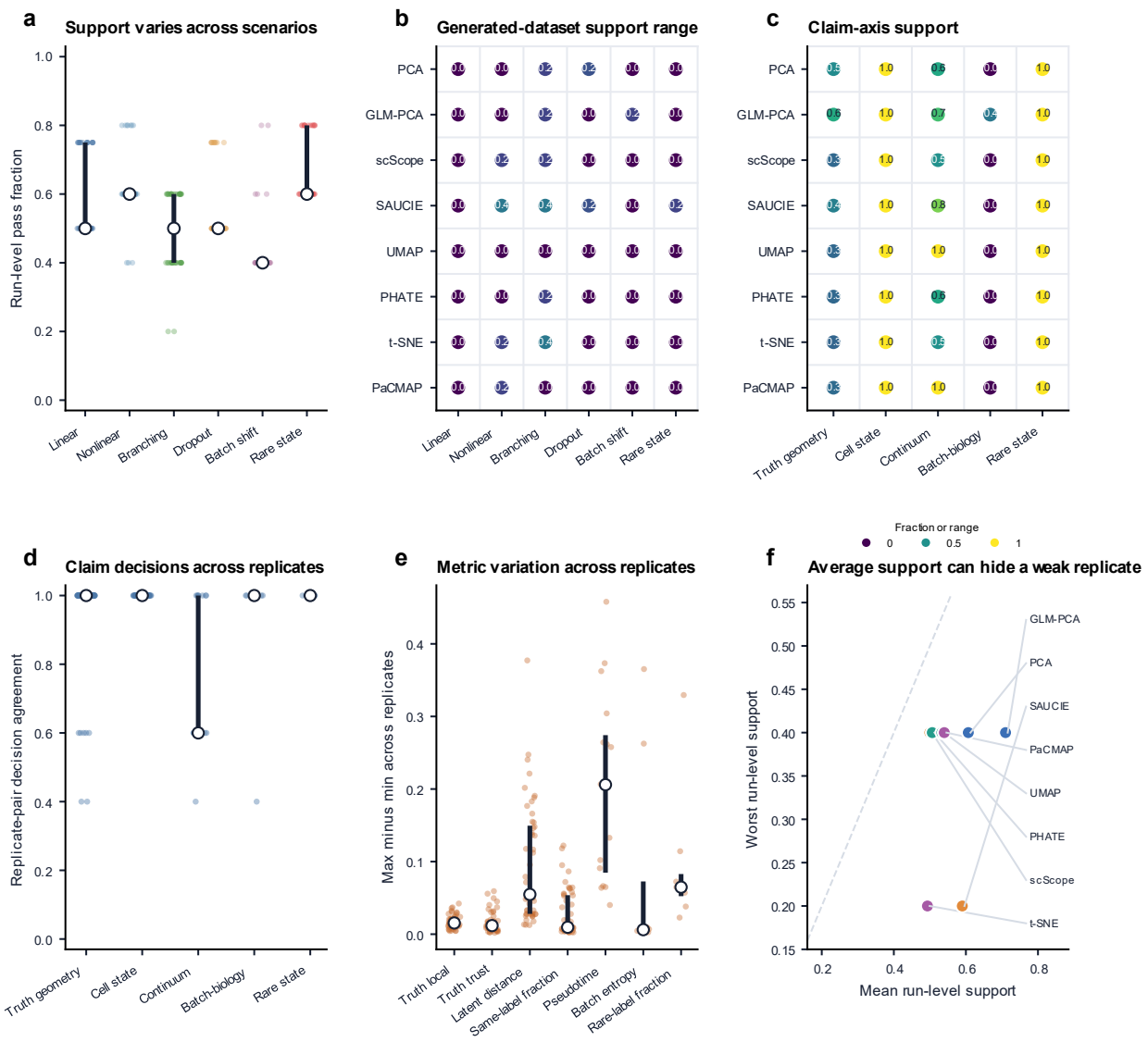

**Supplementary Fig. S10. Local-retention conclusions depend on the reference dimension and neighbourhood size.** A fixed label-stratified subset of 1,000 cells per dataset was used for every calibration. **a,** Mean local retention by method across PCA reference dimensions 2, 5, 10, 20 and 50 at fixed k = 15. **b,** Mean local retention across neighbourhood sizes 5, 15, 30 and 50 at fixed reference dimension 50. **c,** Local retention under PCA2 against PCA50 for matched method-context evaluations; the diagonal marks equality. **d,** Observed local retention at the primary calibration setting against an unrestricted cell-correspondence permutation null. Point colours identify methods and separate points are dataset-specific observations. Grey ticks mark null means and grey bars span central 95% intervals from 100 unrestricted permutations at PCA50 and k = 15; these are null intervals, not confidence intervals. **e,** Number of 24 method-context evaluations below 0.30 for all 20 reference-dimension-by-k definitions. **f,** Spearman concordance of eight-method ranks between PCA reference dimensions. **g,** Context-resolved local retention at PCA50 and k = 15; the dashed line marks 0.30. **h,** Normalised overlap above random expectation, (L - r)/(1 - r), where L is observed local retention and r = 15/999 is random overlap on 1,000 cells; points are context means and whiskers span the observed context range. These calibration results are distinct from the full-object primary analysis. Complete values and permutation definitions are reported in Supplementary Table S10.



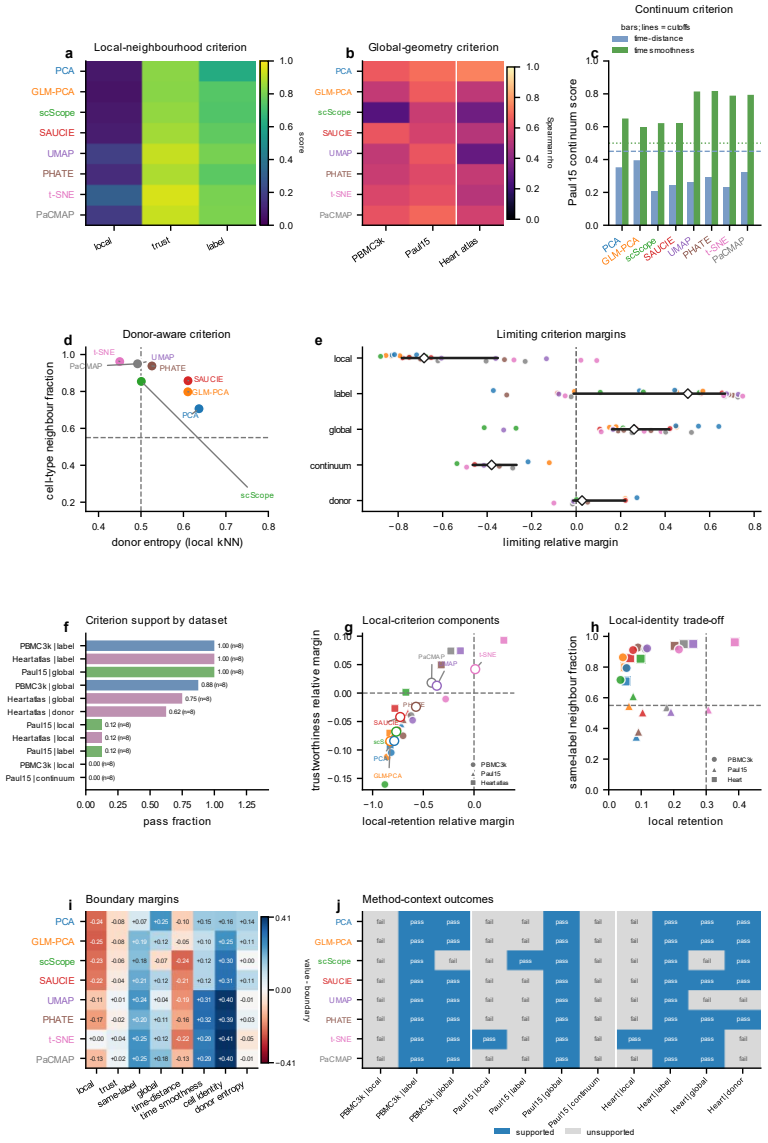

**Supplementary Fig. S11. Paul15 continuum diagnostics are sensitive to diffusion-pseudotime root and lineage.** All analyses use the same 2,730-cell Paul15 object. **a,** All pairwise Spearman correlations of cell pseudotime ranks among the stored 8Mk root and 7MEP, 9GMP and 1Ery centroid roots, displayed as a four-by-four matrix. **b,** Pooled pseudotime-distance correlation for eight methods under each root; the dashed line marks 0.45. **c,** Local pseudotime retention under the same roots; the dashed line marks 0.50. **d,** Within-lineage pseudotime-distance correlation for erythroid, megakaryocytic, basophil and myeloid cluster sets. **e,** Same-branch neighbour fraction by method. **f,** Stored-root pooled correlation against the median lineage-restricted correlation; dashed lines mark 0.45. **g,** Fraction of four roots for which pooled pseudotime-distance correlation met 0.45. **h,** Spearman concordance of eight-method ranks between root pairs. Root choices are alternative analyses of one object and are not independent replicates. Lineage cell counts, sampled-pair denominators and root definitions are reported in Supplementary Table S11.



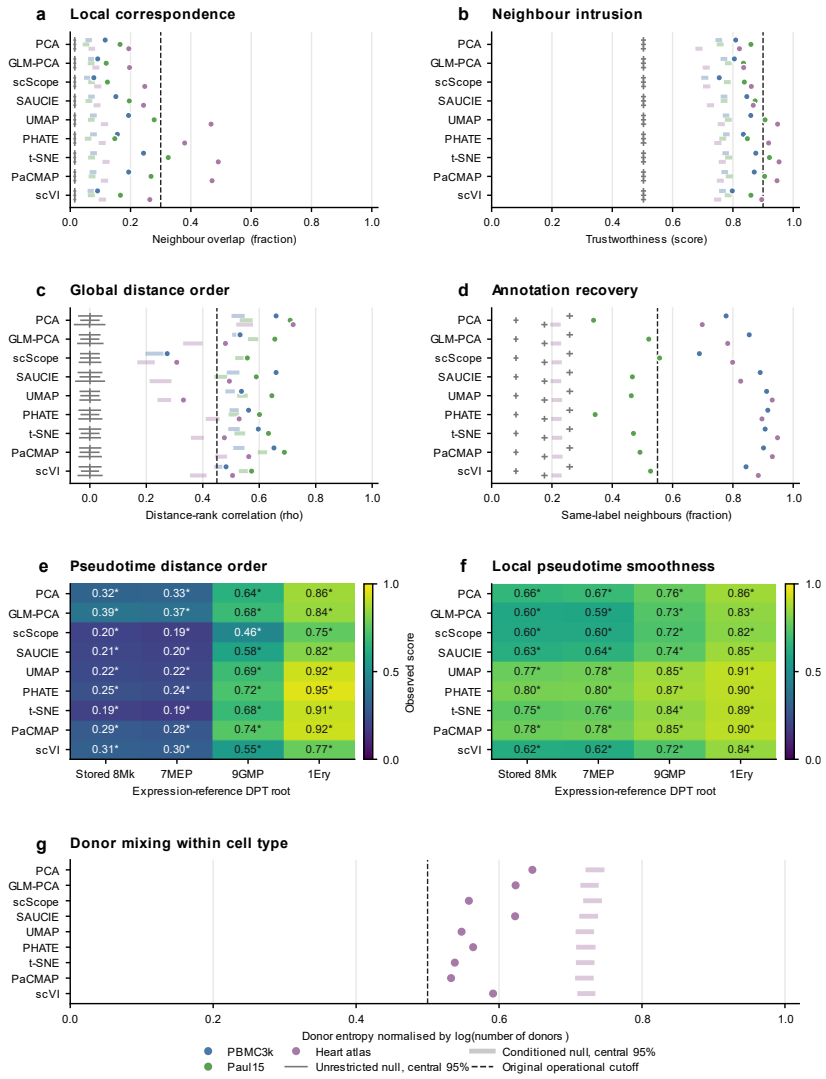

**Supplementary Fig. S12. Matched robustness transitions distinguish unsupported baselines from perturbation-induced support loss. a,** Fractions of matched comparisons classified as pass-to-pass, pass-to-fail, fail-to-pass or fail-to-fail for output dimension, upstream PCA and expression perturbation. **b,** Pass-to-fail fraction among baseline-supported comparisons; labels give lost over eligible comparisons. **c,** Method-specific support loss. Colour encodes the loss fraction; point area increases with the number of baseline-supported comparisons, and labels give lost over baseline-supported comparisons; blank cells are method-axis combinations not evaluated. **d,** Diagnostic-specific support loss using the same encoding. Text marked 0 eligible indicates that no baseline comparison met the boundary. **e,** Pass-to-fail fraction for each non-baseline setting. **f,** Change in signed boundary margin from baseline to condition. Small points are matched comparisons; open circles are medians and vertical bars span the interquartile range. Comparisons are computational conditions on the same analysis objects, not biological replicates. Transition counts and denominators are reported in Supplementary Table S12.



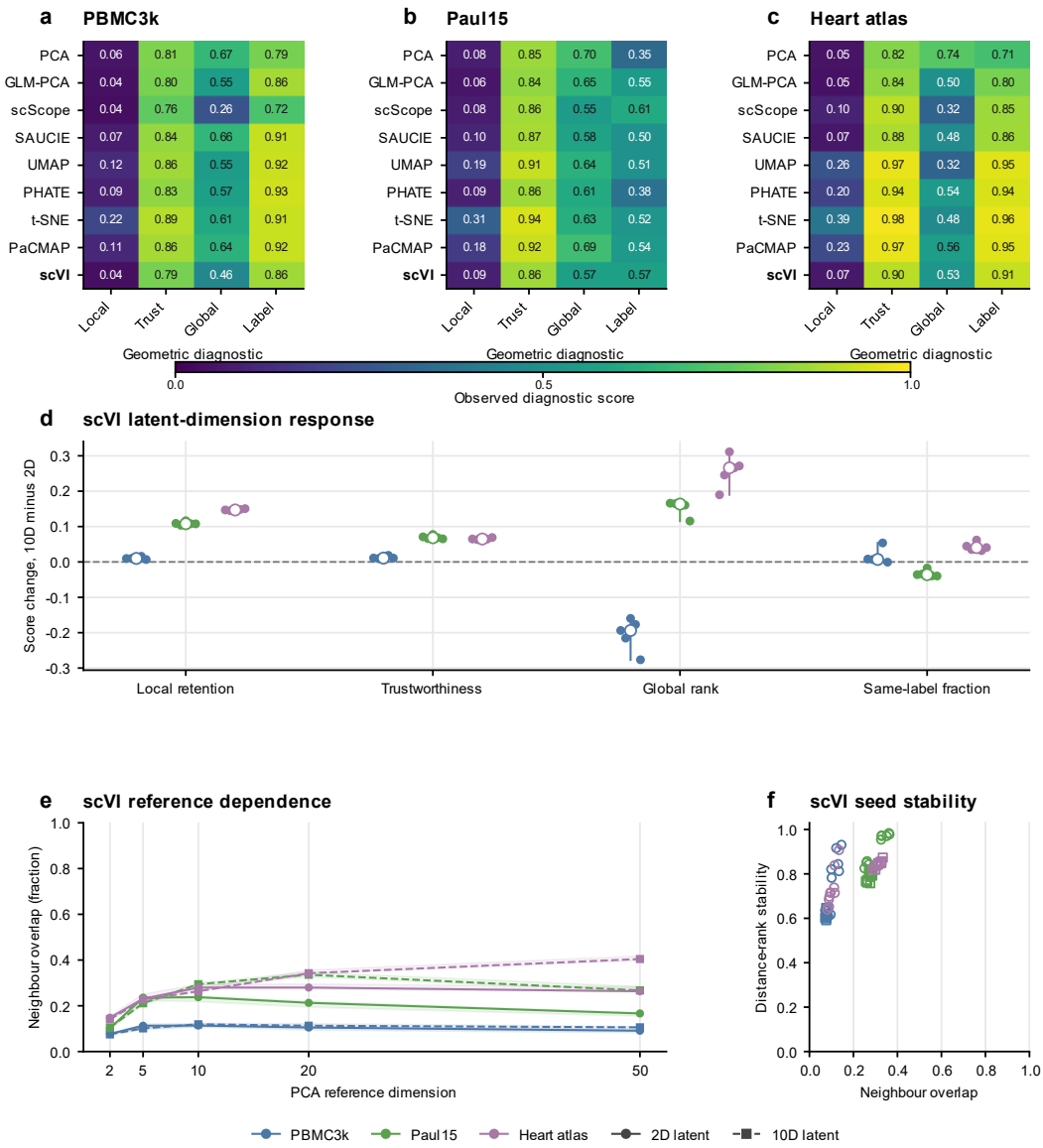

**Supplementary Fig. S13. Five independently generated datasets per scenario quantify controlled stress-test stability. a,** Run-level support fraction across 40 runs for each scenario, comprising eight methods and five generated datasets. Small points are runs; open circles are medians and vertical bars span interquartile ranges. **b,** Range in run-level support across the five generated datasets for every method-scenario combination. In-cell values give the range from 0 to 1. **c,** Interpretation-axis pass fraction for every method across all applicable generated-dataset metric rows. Point colour and label encode the fraction. **d,** Metric-specific decision agreement for the ten shared pairs formed from five generated datasets for generating-structure geometry, cell state, continuum, batch-biology trade-off and rare-state interpretations. Small points are method-scenario-metric agreement summaries, each averaged over ten shared pairs. Open circles are medians and bars span interquartile ranges across these summaries. **e,** Maximum-minus-minimum metric value across generated datasets for each method-scenario-metric combination; points are combinations and open circles are medians and bars span interquartile ranges. **f,** Mean run-level support against worst-run support for each method. The dashed diagonal marks equality, and connectors associate displaced labels with points. Generated datasets are computational replications rather than biological replicates; shared pairs are not independent repetitions. The analysis contains 30 generated datasets, 240 complete runs and 1,120 valid metric rows; complete results are reported in Supplementary Table S13.



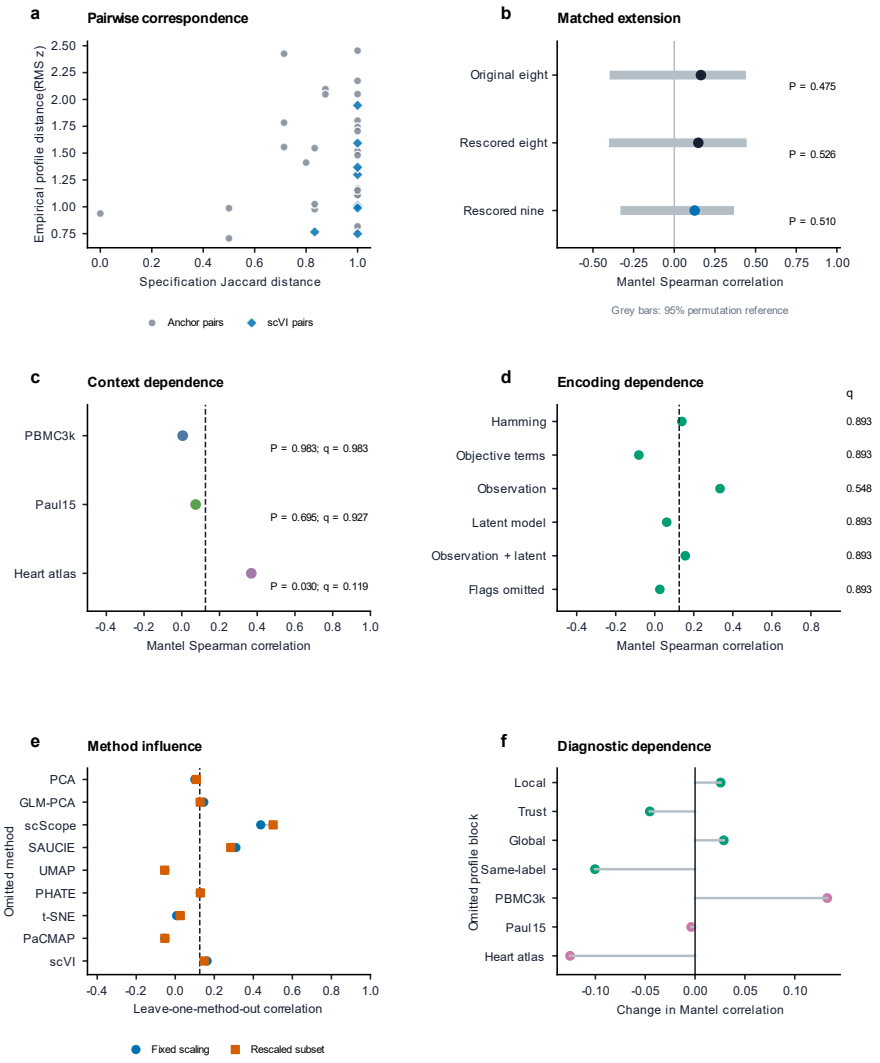

**Supplementary Fig. S14. Operational diagnostic criteria and threshold sensitivity in the eight-method anchor panel. a,** Mean local retention, trustworthiness and same-label neighbour fraction across PBMC3k, Paul15 and the heart subset for each method. **b,** Global-rank correlation by method and dataset. **c,** Paul15 pseudotime-distance correlation and local pseudotime retention under the stored 8Mk-root definition; dashed lines mark operational boundaries of 0.45 and 0.50. **d,** Heart-subset donor entropy and same-cell-type-neighbour fraction; dashed lines mark operational boundaries of 0.50 and 0.55, and the grey connector associates the displaced scScope label with its point. **e,** Limiting signed relative margin for each applicable method-context-claim decision, defined as the smallest component value of (value − boundary)/boundary within a criterion. Coloured points are decisions, black segments span the interquartile range, open diamonds are medians and the dashed line marks zero. **f,** Criterion support by dataset and claim; labels report the supported fraction and number of evaluated method rows. **g,** Joint local-criterion component plane for 24 method-context evaluations. Solid points are individual evaluations, open circles are method means, colour identifies method and symbol identifies dataset. **h,** Local retention against same-label neighbour fraction for the same 24 runs. **i,** Mean observed-minus-boundary margin for each method and diagnostic component. **j,** Binary outcomes for all applicable method-context-claim criteria; blank combinations were not defined and were not counted as failures. The boundaries are study-level operational decision rules, not statistical significance thresholds or universal performance standards. Continuous values and margins are the primary evidence; the binary summaries are retained for sensitivity and auditability.



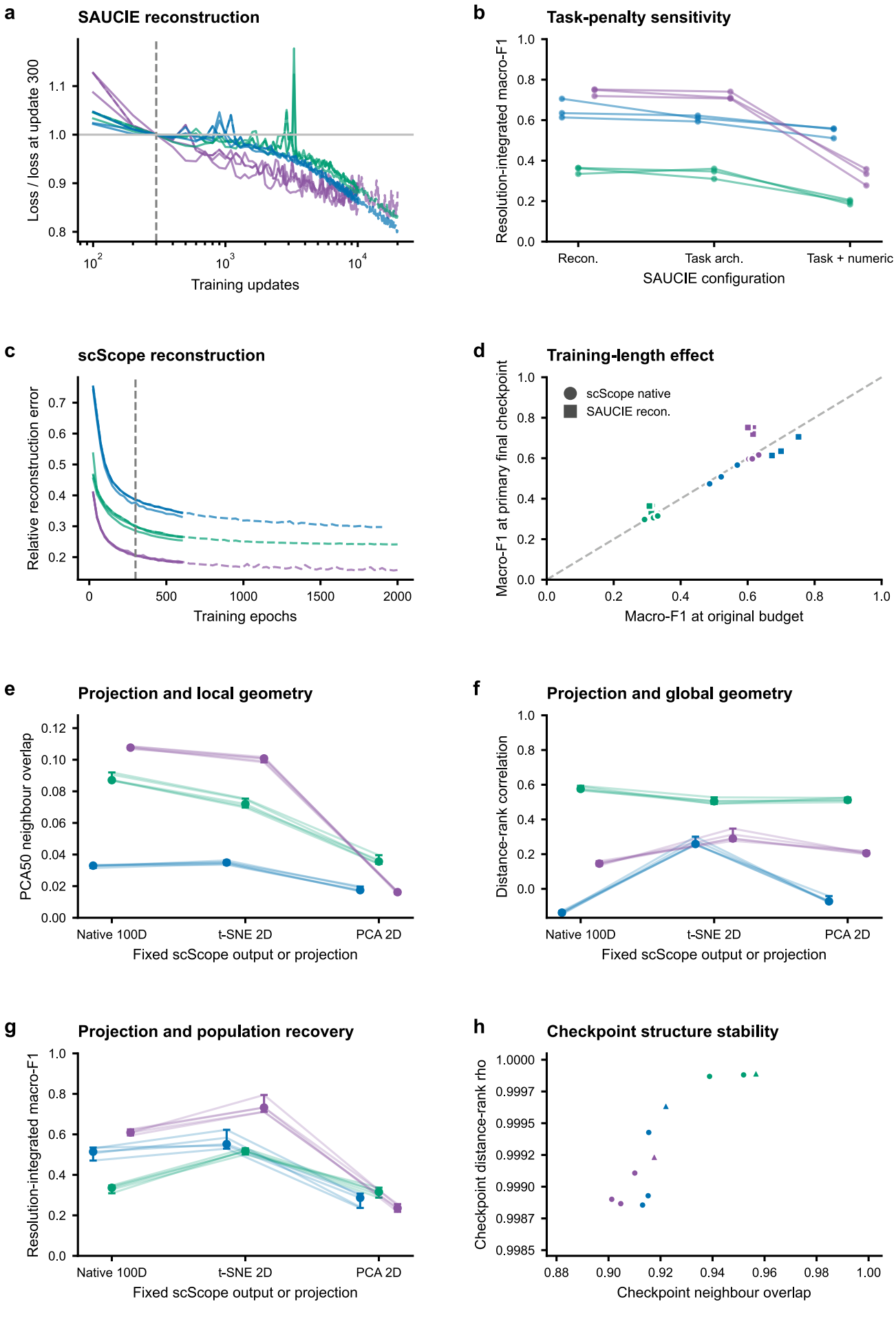

**Supplementary Fig. S15. Random-reference calibration distinguishes non-random structure from operational fidelity. a-d,** Local neighbour overlap, trustworthiness, global distance-rank correlation and same-label neighbour fraction for nine methods on matched 1,000-cell subsets of PBMC3k, Paul15 and heart atlas. Blue, green and purple points identify datasets. Grey bars span central 95% intervals from 1,999 unrestricted permutations; grey ticks mark null means. Coloured translucent bars show annotation-conditioned geometry nulls in a-c and donor-conditioned label permutations for heart in d. Dashed lines mark unchanged operational cutoffs of 0.30, 0.90, 0.45 and 0.55. Local overlap is the shared fraction of 15 nearest neighbours; trustworthiness penalises embedding-only neighbours; global rho compares distance ranks across 5,000 non-self pairs; label fraction measures annotation agreement. **e, f,** Observed pseudotime-distance rho and local pseudotime retention under four expression-reference diffusion pseudotime roots. 8Mk, 7MEP, 9GMP and 1Ery denote megakaryocyte, megakaryocyte-erythroid progenitor, granulocyte-monocyte progenitor and erythroid annotations. Retention is one minus neighbour-to-random-pair mean pseudotime difference. Numbers and colour show observed scores; asterisks denote upper-tail $q < 0.05$ relative to coarse-lineage-conditioned permutations. **g,** Normalised heart donor entropy, with observed points and central 95% intervals from donor permutations within cell types. All nine lower-tail deficit tests have $q = 0.0005$. Donor deficits identify donor association, not technical-batch causality. Probabilities use the plus-one Monte Carlo estimator and separate Benjamini-Hochberg adjustment within each null family. Null intervals are not confidence intervals; cells, pairs and permutations are not biological replicates. scVI is unsupervised and donor-unconditioned. Full effects, probabilities, denominators and cutoff decisions are reported in Supplementary Table S15.



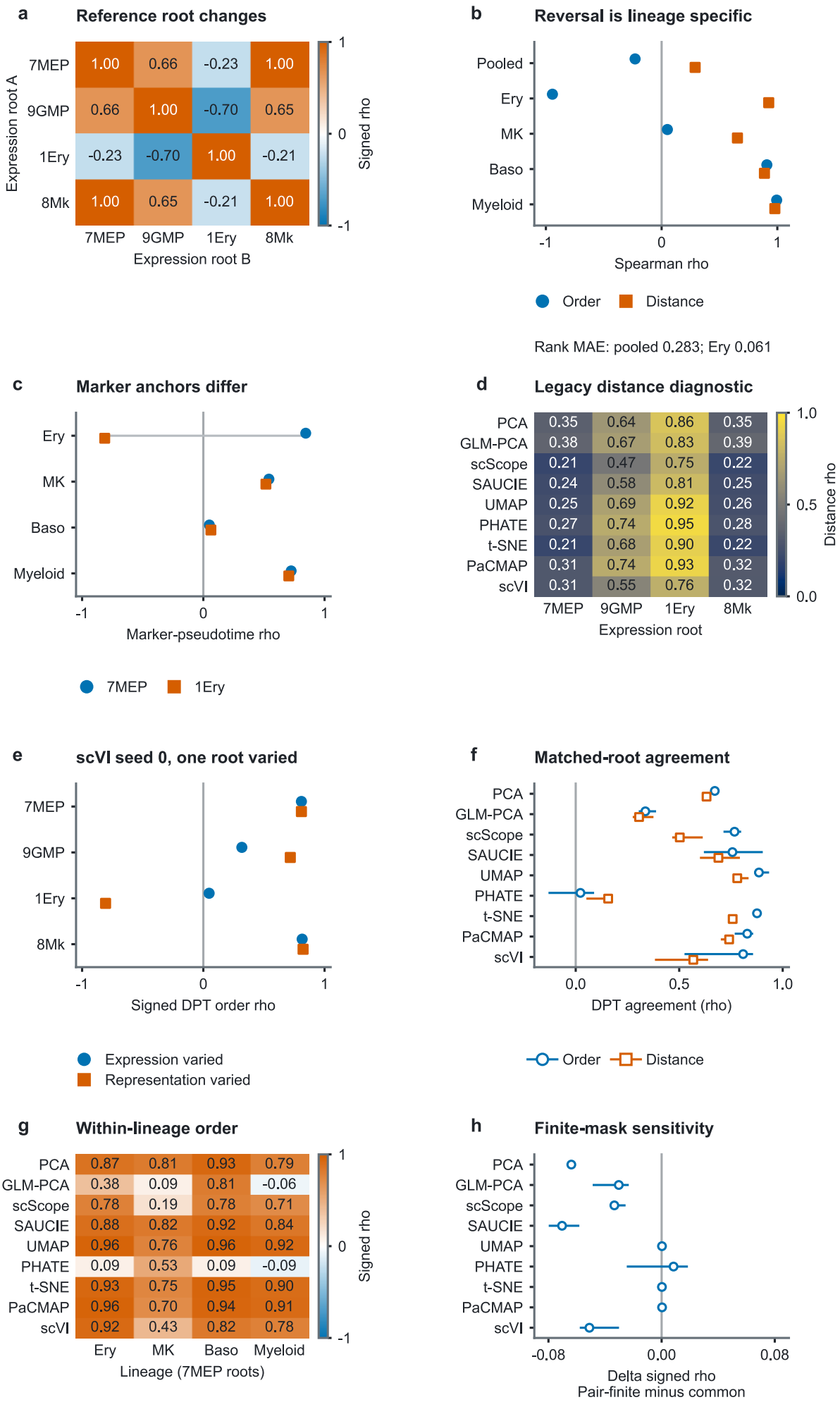

**Supplementary Fig. S16. scVI geometry depends on biological context and reference definition. a-c,** Four geometric diagnostics for the eight anchor methods and scVI at output dimension two and seed zero in PBMC3k, Paul15 and heart atlas. Local, Trust, Global and Label denote reference-neighbour overlap, trustworthiness, Spearman distance-rank correlation and same-label neighbour fraction. Colour and numbers show observed scores; the common scale is zero to one. The full objects contain 2,638, 2,730 and 8,000 cells. Neighbour size is 15, the reference is PCA50 and global scores use 5,000 fixed non-self cell pairs with sampling seed 20261005. **d,** Paired score changes between ten- and two-dimensional scVI latents. Small points denote five seeds, open circles medians and vertical bars observed ranges; the horizontal line marks no change. **e,** Local overlap across reference PCA dimensions on matched 1,000-cell subsets. Lines show medians over five seeds and shading their observed ranges. Solid circles denote two-dimensional latents and dashed squares ten-dimensional latents. **f,** Neighbour overlap against distance-rank agreement between independently seeded scVI outputs. Each context and dimension contributes ten seed-pair comparisons formed from five seeds; circles denote two-dimensional and squares ten-dimensional latents. Blue, green and purple denote PBMC3k, Paul15 and heart atlas throughout d-f. scVI was fitted without annotations or donor covariates, so these panels do not evaluate batch-conditioned scVI. Ranges describe computational variation, not confidence intervals. Cells, sampled pairs, seeds and shared seed pairs are not independent biological replicates. No inferential tests or significance symbols are used. Full diagnostic, reference-sensitivity and stability values are reported in Supplementary Table S16.




**Supplementary Fig. S17. scVI inclusion does not produce a detected pooled specification-profile association.** Nine seed-zero two-dimensional representations were scored in PBMC3k, Paul15 and heart subsets containing 2,638, 2,730 and 8,000 cells, using PCA50, k = 15 and 5,000 fixed non-self pairs. a, Jaccard distances between 20-feature specifications versus empirical-profile distances. Grey circles are 28 anchor pairs; blue diamonds are eight scVI pairs. RMS z is Euclidean distance divided by the square root of 12 column-standardised variables. Pairs are dependent; n = 9 methods. b, Mantel correlations for original eight-method, rescored eight-method and rescored nine-method analyses. Grey bars show central 95% exact permutation references, not confidence intervals. Controls distinguish rescoring from inclusion; differences between correlations were not tested. c, Dataset-specific four-variable profiles; blue, green and purple indicate PBMC3k, Paul15 and heart. Dashed lines mark the pooled correlation in c and e and the primary Jaccard result in d. P denotes exact two-sided probabilities; q applies Benjamini-Hochberg correction across the pooled and three context tests. d, Six alternative encodings with a separate six-test correction. Green dots show correlations; the right column gives q. Flags omitted excludes not-explicit indicators, not measurements. e, Method omissions with fixed full-nine scaling (blue circles) or subset rescaling (orange squares); grey segments connect paired results. f, Correlation changes after diagnostic (green) or dataset (pink) omission. Local, Trust, Global and Same-label denote neighbour overlap, trustworthiness, distance-rank correlation and same-label neighbour fraction. e and f are descriptive influence analyses. Tests enumerate all 40,320 or 362,880 whole-method assignments, including identity, with average ranks for ties. This post hoc analysis does not infer biological replication or population-wide effects. Supplementary Tables S16 and S17 provide measurements and statistical results.

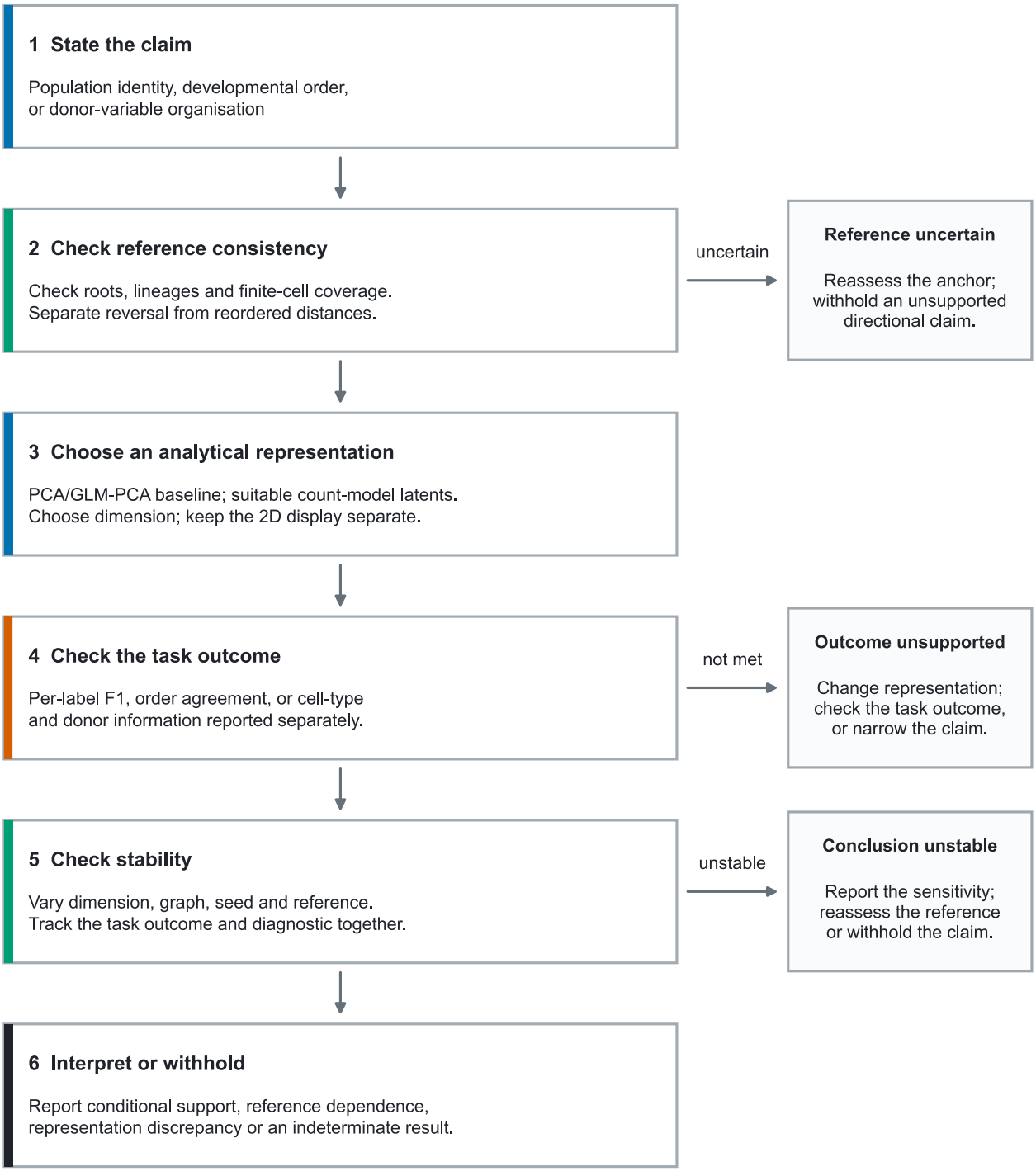

a
SAUCIE reconstruction
Loss / loss at update 300
Training updates
b
Task-penalty sensitivity
Resolution-integrated macro-F1
Recon.
Task arch.
Task + numeric
SAUCIE configuration
c
scScope reconstruction
Relative reconstruction error
Training epochs
d
Training-length effect
scScope native
SAUCIE recon.
Macro-F1 at primary final checkpoint
Macro-F1 at original budget
e
Projection and local geometry
PCA50 neighbour overlap
Native 100D
t-SNE 2D
PCA 2D
Fixed scScope output or projection
f
Projection and global geometry
Distance-rank correlation
Native 100D
t-SNE 2D
PCA 2D
Fixed scScope output or projection
g
Projection and population recovery
Resolution-integrated macro-F1
Native 100D
t-SNE 2D
PCA 2D
Fixed scScope output or projection
h
Checkpoint structure stability
Checkpoint distance-rank rho
Checkpoint neighbour overlap
PBMC3k
Paul15
Heart atlas

**Supplementary Fig. S18. Training, projection and active configuration alter deep-workflow interpretation.** All panels use PBMC3k, Paul15 and heart analysis objects containing 2,638, 2,730 and 8,000 cells. Blue, green and purple identify these datasets. a, Genuine reconstruction-only SAUCIE losses divided by their update-300 value; each line is a fit. Solid curves use the primary control caps; dashed curves are separately labelled longer-budget fits. The vertical line marks 300 updates. b, Matched final population recovery for the default reconstruction branch, task-architecture reconstruction control and task-penalty numerical sensitivity. Lines connect the same dataset and seed. Enabling task penalties also changes activations; the middle control separates that architectural change. The task sensitivity explicitly enforces nonnegative finite squared distances before the original kernel and square root; unmodified task code produced nonfinite loss and is not silently replaced. c, Original scScope masked relative reconstruction errors; the vertical reference marks 300 epochs, and dashed curves denote longer-budget fits. d, Resolution-integrated macro-F1 at the original versus primary final budget for the same new fits; circles denote native scScope and squares reconstruction-only SAUCIE. The diagonal marks equality. e-g, Matched projection controls for saved 100-dimensional scScope predictions and their two-dimensional t-SNE and PCA projections, assessing PCA50 neighbour overlap, global distance-rank correlation and annotated-population macro-F1 separately. Thin lines connect individual seeds; points show medians and whiskers the full observed range. h, Neighbour overlap and distance-rank concordance between the last preterminal 25-epoch checkpoint and final native scScope output. Circles use the primary cap and triangles the longer budget. Primary controls use three computational seeds per dataset; e-g use five original seeds. The six longer-budget fits use seed zero only and are not additional independent seeds. Macro-F1 is the equal-label average of the harmonic mean of precision and recall after unsupervised Leiden clustering and post hoc majority mapping, integrated over the fixed resolution grid. Neighbour size is 15 and global scores use 5,000 fixed nonself pairs. Trustworthiness and additional population measurements are recorded in Supplementary Table S18. Primary caps are 10,000 SAUCIE updates and 600 scScope epochs; the loss-driven extension caps are 20,000 updates and 2,000 epochs. Early stopping requires three successive relative loss changes below 0.001 after the minimum budget. The strict loss rule was met by 0 of 36 primary fits and 1 of six extensions. Unmet rules remain unmet; these plots do not establish optimum training or biological reliability. Seed ranges are computational variation, not confidence intervals; no biological significance tests are applied.


a
Reference root changes
Expression root A
7MEP
9GMP
1Ery
8Mk
1.00 0.66 -0.23 1.00
0.66 1.00 -0.70 0.65
-0.23 -0.70 1.00 -0.21
1.00 0.65 -0.21 1.00
7MEP 9GMP 1Ery 8Mk
Expression root B
Signed rho
1
0
-1
b
Reversal is lineage specific
Pooled
Ery
MK
Baso
Myeloid
-1
0
1
Spearman rho
Order
Distance
Rank MAE: pooled 0.283; Ery 0.061
c
Marker anchors differ
Ery
MK
Baso
Myeloid
-1
0
1
Marker-pseudotime rho
7MEP
1Ery
d
Legacy distance diagnostic
PCA 0.35 0.64 0.86 0.35
GLM-PCA 0.38 0.67 0.83 0.39
scScope 0.21 0.47 0.75 0.22
SAUCIE 0.24 0.58 0.81 0.25
UMAP 0.25 0.69 0.92 0.26
PHATE 0.27 0.74 0.95 0.28
t-SNE 0.21 0.68 0.90 0.22
PaCMAP 0.31 0.74 0.93 0.32
scVI 0.31 0.55 0.76 0.32
7MEP 9GMP 1Ery 8Mk
Expression root
Distance rho
1.0
0.5
0.0
e
scVI seed 0, one root varied
7MEP
9GMP
1Ery
8Mk
-1
0
1
Signed DPT order rho
Expression varied
Representation varied
f
Matched-root agreement
PCA
GLM-PCA
scScope
SAUCIE
UMAP
PHATE
t-SNE
PaCMAP
scVI
0.0
0.5
1.0
DPT agreement (rho)
Order
Distance
g
Within-lineage order
PCA 0.87 0.81 0.93 0.79
GLM-PCA 0.38 0.09 0.81 -0.06
scScope 0.78 0.19 0.78 0.71
SAUCIE 0.88 0.82 0.92 0.84
UMAP 0.96 0.76 0.96 0.92
PHATE 0.09 0.53 0.09 -0.09
t-SNE 0.93 0.75 0.95 0.90
PaCMAP 0.96 0.70 0.94 0.91
scVI 0.92 0.43 0.82 0.78
Ery MK Baso Myeloid
Lineage (7MEP roots)
Signed rho
1
0
-1
h
Finite-mask sensitivity
PCA
GLM-PCA
scScope
SAUCIE
UMAP
PHATE
t-SNE
PaCMAP
scVI
-0.08
0.00
0.08
Delta signed rho
Pair-finite minus common

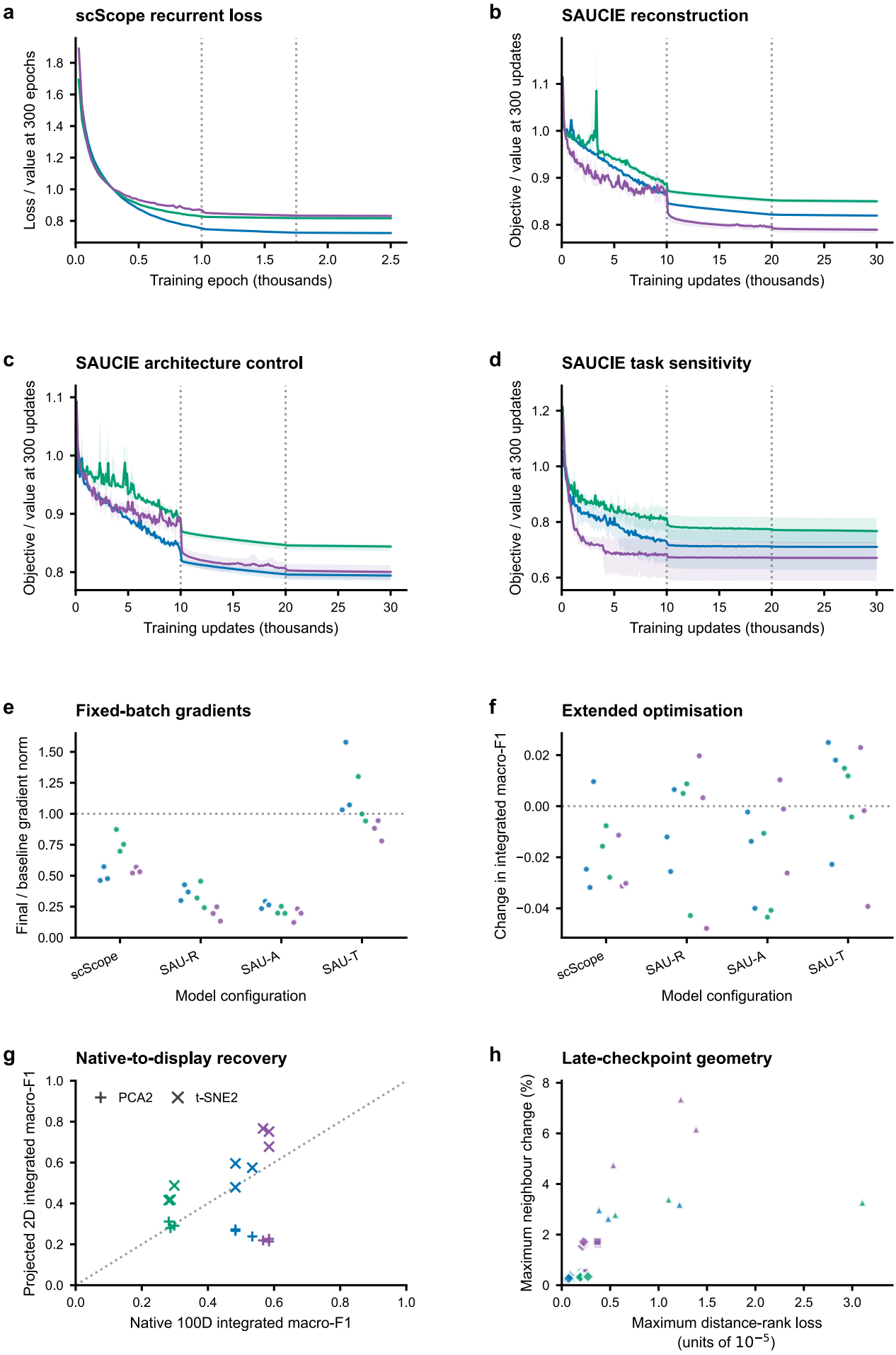
**Supplementary Fig. S19. Reference-attribution sensitivity.** a, Signed Spearman correlation between expression-reference diffusion pseudotimes under four roots. b, Reference change from 7MEP to 1Ery, pooled and lineage-restricted. Blue circles show signed order correlation; orange squares show correlation of absolute pseudotime differences across fixed cell-pair draws, which is reversal invariant. Rank MAE is the smaller mean absolute error between direct or reversed average-tie ranks scaled to [0,1]. Erythroid order is largely, but not exactly, reversed; the pooled change is not a global reversal. c, Lineage marker-expression associations with reference pseudotime under 7MEP (blue circles) and 1Ery (orange squares); grey connectors pair programmes, not trajectories. Marker scores are mean log1p expression and provide same-expression consistency, not independent validation. d, Separate legacy diagnostic correlating absolute reference-pseudotime differences with fixed Euclidean embedding distances, not representation-derived pseudotime. e, Actual scVI seed-zero comparison. Blue circles vary the expression root with representation root fixed at 7MEP; orange squares vary the representation root with expression root fixed at 7MEP. Graphs and coordinates remain fixed; both 7MEP symbols repeat one baseline. f, Matched-7MEP expression-to-representation pseudotime agreement. Open blue circles show signed order correlation; open orange squares show absolute-distance correlation. Points are five-seed medians; bars span the full observed computational range. g, Median matched-7MEP signed order agreement within each lineage; scalar pseudotime does not measure branch topology. h, Signed-correlation change, method-pair-finite minus global-common, at matched 7MEP roots. Points and bars show seed medians and full ranges. Maximum absolute change is 0.079035. The common mask intersects all roots, nine methods and five seeds. Panels a-c and e-h use exactly 2,670 common finite cells of 2,730, excluding 60. Common lineage intersections are erythroid 1,262, megakaryocytic 235, basophil 535 and myeloid 970; masks overlap. Panel d uses all 2,730 cells; h additionally uses 2,672-2,730 method-pair-finite cells. Ery, MK and Baso denote the first three lineages. Root labels are annotation identifiers: 7MEP, megakaryocyte-erythroid progenitor; 9GMP, granulocyte-monocyte progenitor; 1Ery, erythroid; 8Mk, stored megakaryocyte root. DPT means diffusion pseudotime. Heatmaps show dimensionless correlations, [-1,1] in a and g and [0,1] in d. Grey zero lines indicate zero correlation/change, not thresholds. Seeds 0-4 are computational repeats, not biological replicates; PCA and evaluated t-SNE coordinates repeat across seed labels. Ranges are not confidence intervals. Undefined pseudotimes remain missing. All analyses are descriptive; no cutoff is applied to crossed DPT. Full graph settings and fixed pair-draw definitions are in Supplementary Table S19. Panel-level source records accompany the figure.

Conceptual model; not a quantitative measurement.




No fitted result, method ranking or universal reliability threshold is shown.

**Supplementary Fig. S20. Conceptual decision map for single-cell analysis, not an empirical result.** Conceptual model; not a quantitative measurement. The six central steps connect a stated single-cell analysis claim to checks of reference consistency, selection of an analytical representation, evaluation of the task outcome, stability checks and interpretation or withholding of the claim. Reference checking includes root, lineage and finite-cell checks and distinguishes orientation reversal from changed relative distances. Representations are chosen in view of the observation model and output dimension; a two-dimensional display is kept separate from the representation used for inference. PCA, principal component analysis, and GLM-PCA, generalized linear model principal component analysis, are illustrative baselines, while count-model latents are candidates only when their assumptions fit the data. The diagram does not identify a universally best method. Measures of the task outcome include per-population F1, the harmonic mean of precision and recall; agreement of inferred developmental order with a stated reference; or cell-type and donor information evaluated separately. Stability checks examine the task outcome together with its diagnostic rather than rely on stable appearance. Right-hand branches indicate actions when the reference is uncertain, the task outcome is unsupported or the conclusion changes under plausible alternatives. Downward arrows denote continuation within a stated scope; horizontal arrows denote an alternative or stop decision, not measured transitions or probabilities. Coloured strips distinguish reasoning stages, not performance levels. Reference-dependent, representation-dependent and indeterminate interpretations are conditional categories, not validated diagnoses of biological truth. The schematic contains no observed scores, sample sizes, uncertainty intervals or statistical comparisons. It complements, and does not replace, the actionable task routes in main Table 2. Diagram nodes and connections accompany the figure as conceptual source records.


a
scScope recurrent loss
Loss / value at 300 epochs
Training epoch (thousands)
b
SAUCIE reconstruction
Objective / value at 300 updates
Training updates (thousands)
c
SAUCIE architecture control
Objective / value at 300 updates
Training updates (thousands)
d
SAUCIE task sensitivity
Objective / value at 300 updates
Training updates (thousands)
e
Fixed-batch gradients
Final / baseline gradient norm
scScope
SAU-R
SAU-A
SAU-T
Model configuration
f
Extended optimisation
Change in integrated macro-F1
scScope
SAU-R
SAU-A
SAU-T
Model configuration
g
Native-to-display recovery
PCA2
t-SNE2
Projected 2D integrated macro-F1
Native 100D integrated macro-F1
h
Late-checkpoint geometry
Maximum neighbour change (%)
Maximum distance-rank loss
(units of $10^{-5}$)
PBMC3k
Paul15
Heart atlas
scScope
SAU-R
SAU-A
SAU-T

**Supplementary Fig. S21. Fixed-budget optimisation and representation controls.** All panels use measured fits from PBMC3k, Paul15 and the heart-atlas subset, with 2,638, 2,730 and 8,000 cells, respectively. Colours denote datasets. Three indexed seeds per dataset and configuration are computational repeats, not biological replicates; there are 36 fitted models and 90 evaluated representation endpoints. SAU-R denotes the reconstruction configuration, SAU-A denotes reconstruction under the task-branch architecture, and SAU-T denotes the task-enabled numerical sensitivity, with finite negative squared distances projected to zero as described in Methods. The unmodified task-code failure remains separate and is not represented as a fit. a, Full scScope relative reconstruction monitor, summing the masked Frobenius residual norms of both recurrent outputs and dividing by the norm of the same observed input. This diagnostic is distinct from the last-recurrence monitor and from the published per-cell squared formulation. b-d, SAUCIE total diagnostic objectives for reconstruction, architecture-matched reconstruction and the task-enabled numerical sensitivity. Objectives are sample-size-weighted means across the same four fixed 256-cell training groups, not joint 1,024-cell task losses. In a-d, each seed trajectory is normalised by its own value at epoch 300 or update 300; lines show three-seed means and shaded bands show observed minima and maxima, not confidence intervals. Dotted vertical lines mark the two predefined learning-rate reductions. The horizontal coordinate denotes actual epochs or updates in thousands. Curves with different active terms are not comparable as absolute objective values. e, Ratio of terminal to baseline Euclidean gradient norm on the same fixed 64-cell scScope or 256-cell SAUCIE batch. Baselines are epoch 600 and update 10,000, respectively. Each point is one fit, displaced horizontally only for readability; the dotted line denotes no change. A smaller norm is a training diagnostic, not evidence of biological validity or a global optimum. f, Paired terminal minus baseline resolution-integrated macro-F1 for each fit. The eight Leiden resolutions span 0.2-2.0; integration is trapezoidal and divided by the span of 1.8. Majority label mapping is performed only after fitting and clustering. The dotted line denotes no endpoint change. g, Terminal native 100-dimensional scScope recovery compared with PCA and t-SNE projections of the identical saved coordinates; plus and cross symbols distinguish the projections. The diagonal denotes equal integrated macro-F1 and there are nine matched fits per projection. h, For each fit, the maximum distance-rank loss, defined as one minus Spearman distance-rank agreement, and maximum percentage neighbour change, defined as 100 times one minus 15-neighbour overlap, across consecutive retained late coordinate checkpoints. The horizontal scale is in units of 0.00001. These maxima are marginal summaries and need not arise from the same pair. Symbols denote configurations. Late windows begin at epoch 2,000 or update 25,000; rank agreement uses 5,000 fixed nonself cell pairs and overlap uses all cells. No biological hypothesis test, confidence interval or significance marker is assigned. Recorded loss components, endpoints and read-only saved-state gradient checks are in Supplementary Table S20. Detailed scoring and random streams are specified in Methods.